

Fine mass scans, injected components, and shared peak separations

Standalone continuation of the v5.2.0 study

Version 5.2.1 — 10 September 2026

Main findings

The fine scan resolves the shape of the 2016 upper-limit trough, while the fitted two-peak spectrum reproduces much of its depth and width. In the fixed 100–110 MeV interval, the observed minimum is at 101 MeV, $\epsilon_{90}^2 = 1.55 \times 10^{-6}$; the original 90/117 MeV fitted mean predicts its minimum at 102.5 MeV and 1.78×10^{-6} . The half-MeV scan therefore does not move the observed integer-grid minimum, and it makes the remaining offset visible.

The expanded targeted pair catalogue selects 66/93 MeV in the all-three search, with an improvement over the best single peak of $\Delta_{2/1} = 5.47$. This comparison includes the exploratory extension of 2015 to 100 MeV. The strongest individual-scope pair remains 82/91 MeV in 2016. Neither is a calibrated global significance.

Two injected components at 90/117 MeV reproduce the 2016 limit shape better than the fitted continuum alone. Adding a positive 65 MeV component improves the 2021 and combined shape comparisons. These are post-fit morphology experiments: their continuum and reference strengths use the same observed data, and ten trials per condition cannot establish coverage or a discovery probability.

Allowing a shared separation with year-dependent offsets changes the all-three 80 MeV-family likelihood only slightly: $2\Delta \log L = 0.34$. The fitted separation is 12.25 MeV, with offsets 0, +1.88, and +2.81 MeV for 2015, 2016, and 2021. Releasing the shared-strength assumption gives a much larger gain of 43.11. Similar spacing alone does not establish compatible production rates.

Scope of the continuation

Calculation	Completed
Observed half-MeV limits	1,433 fits in seven searches
Targeted pair catalogue	2,968 screening fits; 600 broad comparisons
Poisson injection pilot	10 trials per scheme and strength
Injection conditions	2 schemes $\times$ 5 strengths $\times$ 10 trials = 100 labels
Shared-separation comparison	36 mass/rate models; 30 profile points

The original v5.2.0 PDF is preserved. The accompanying source package contains the spectra, fixed states, complete numerical tables, every displayed extraction, all ten toy curves per condition, and reproducible scripts. The regions and extensions were chosen after viewing the data; this report develops and compares possible explanations rather than testing a predeclared discovery hypothesis.

1. Count model, half-MeV limits, and the 2015 extension

The inputs remain the full 2015 and 2016 spectra and the released native 2021 10% sample. Analysis-bin widths are 0.25, 0.25, and 0.625 MeV. The hypothesis domains are 19–100, 39–180, and 50–250 MeV; GP supports remain 14–135, 30–210, and 36–300 MeV. Combinations use their common mass domains. No finer histogram or new data increment is introduced.

At each half-MeV hypothesis the original moving $\pm 2.25\sigma$ interval is excluded from GP training. Positive kernel coordinates are interpolated linearly in their logarithms between archived integer states; integer coordinates are retained exactly. Beyond 90 MeV, 2015 uses its frozen 90 MeV state. This extends the hypothesis domain without qualifying new support or coverage. Nearest-state, fixed-91 MeV, and frozen-85 MeV endpoint controls are displayed separately.

Write $A = \epsilon^2/10^{-8}$ and let S be the full-yield, bin-integrated signal template in counts per unit A . The mean and objective are

$$\lambda = b + L\theta + AS, \quad \ell = \sum_i [\lambda_i - n_i + n_i \log(n_i/\lambda_i)] + \frac{1}{2}\theta^T \theta.$$

For zero counts, the Poisson contribution is λ_i . The GP nuisance covariance is LL^T . Each combined fit stacks the campaign likelihoods with independent GP nuisances and one shared A . The retained resolution and observed-density conversion remain fixed; their uncertainties are not newly profiled.

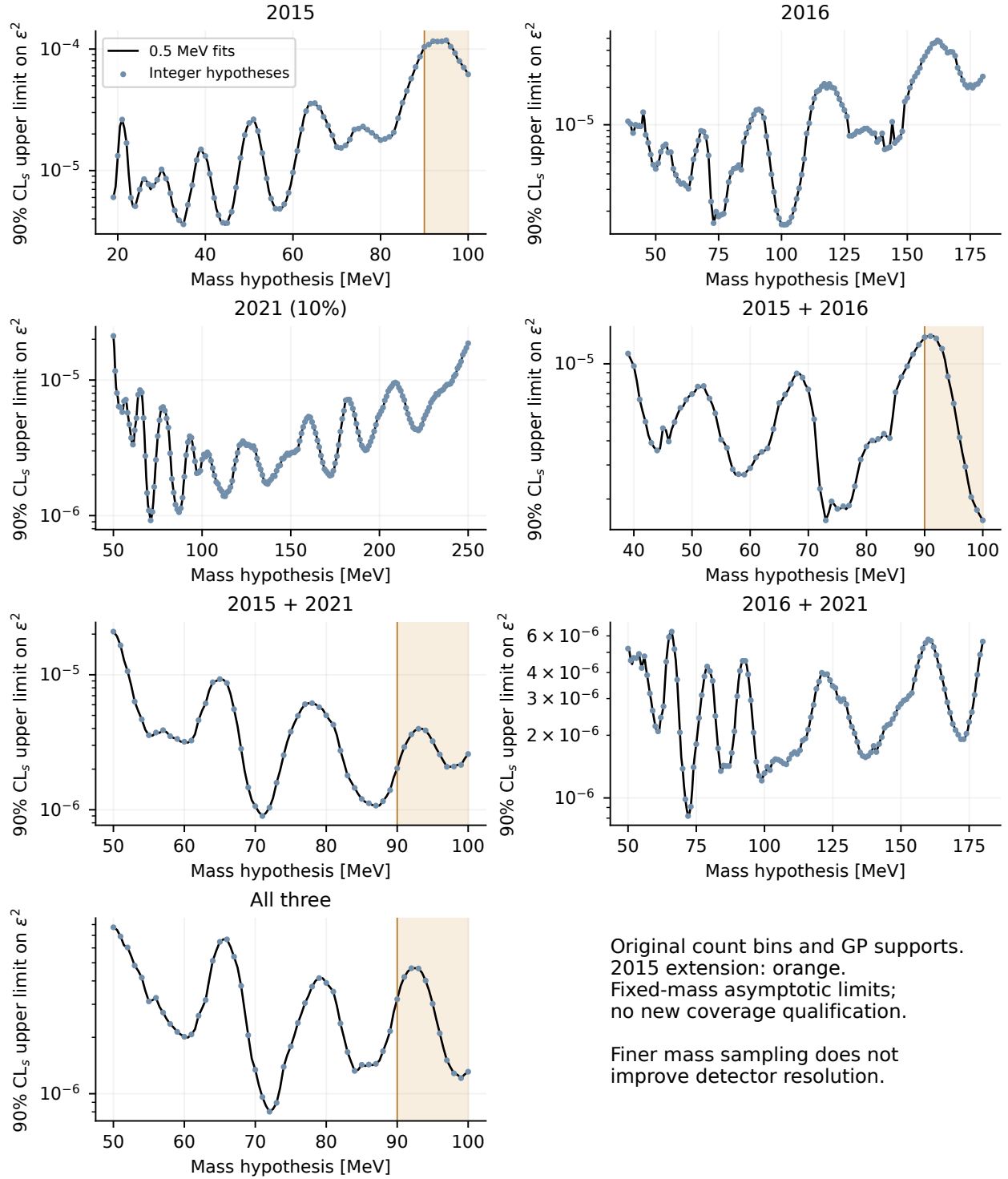
The upper limit solves $\text{CL}_s(A) = 0.1$ using explicitly profiled observed and background-Asimov likelihood ratios, rather than a Gaussian estimate of the limit. For $q = \tilde{q}_A > 0$ and $q_A^{\text{Asimov}} = q_a$, the bounded asymptotic tails [4] use

$$(z_{s+b}, z_b) = \begin{cases} (\sqrt{q}, \sqrt{q} - \sqrt{q_a}), & q \leq q_a, \\ ((q + q_a)/(2\sqrt{q_a}), (q - q_a)/(2\sqrt{q_a})), & q > q_a, \end{cases} \quad \text{CL}_s = \frac{\Phi(-z_{s+b})}{\Phi(-z_b)}.$$

The denominator of the observed likelihood ratio uses the physical nonnegative fit; the upper-tail statistic is zero when the fitted strength exceeds the tested strength. The second branch matters in deficits. The Asimov mean and Gaussian constraint are those of the fixed mass-specific model; this is a conditional asymptotic calculation, not a newly calibrated sampling distribution.

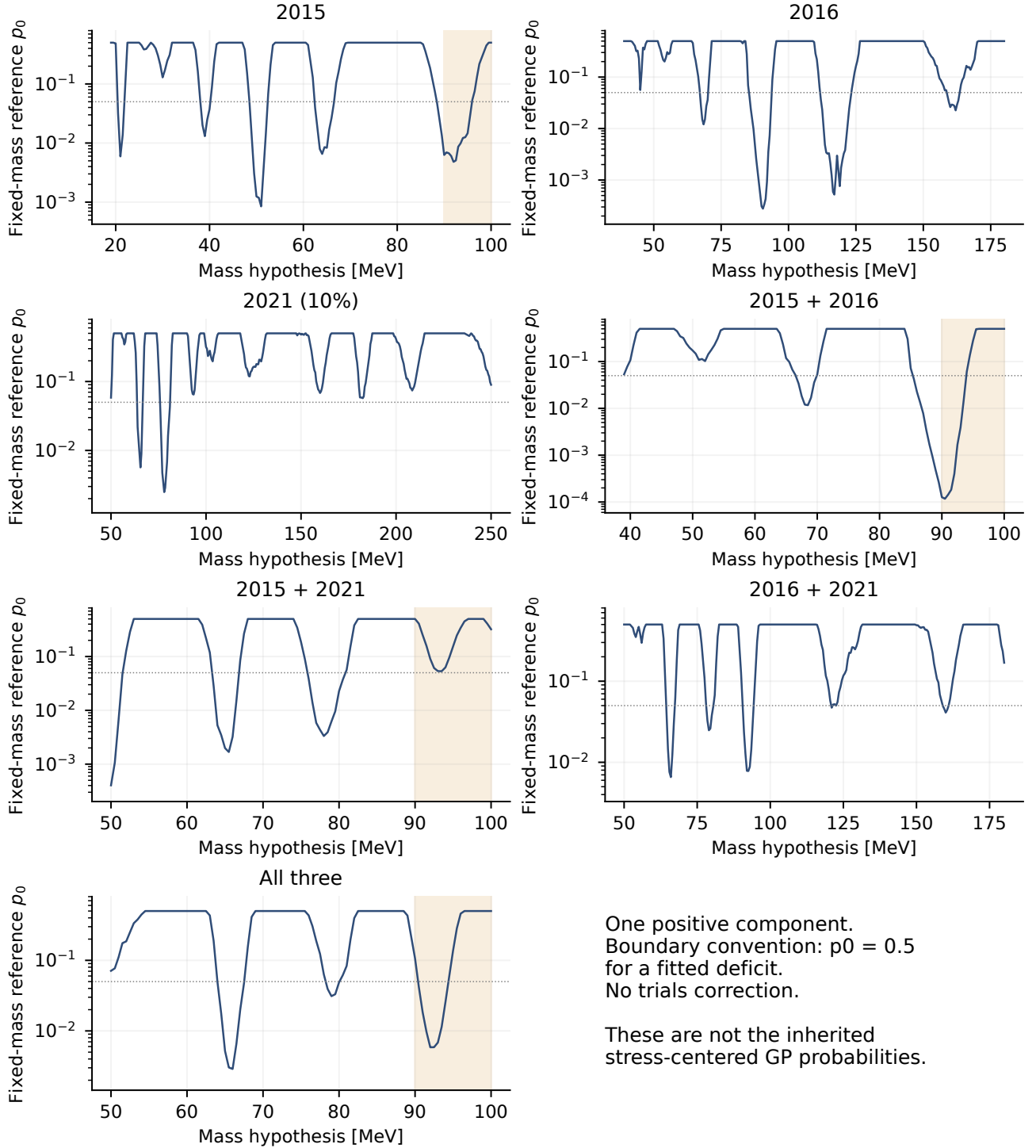
A half-MeV step samples the response more closely than a one-MeV step. It does not improve detector resolution, eliminate binning effects, or supply a trials correction. The inherited 2016 numerical/qualification exception and fixed-normalization assumptions remain relevant [2,3].

2. Observed upper-limit curves



Actual bounded 90% CL_s solutions at 0.5 MeV spacing. Small blue points show the integer hypotheses. Orange marks the 2015 extension, including combinations that retain 2015. Each curve uses its own appropriate campaign likelihood; overlapping combinations are not independent experiments.

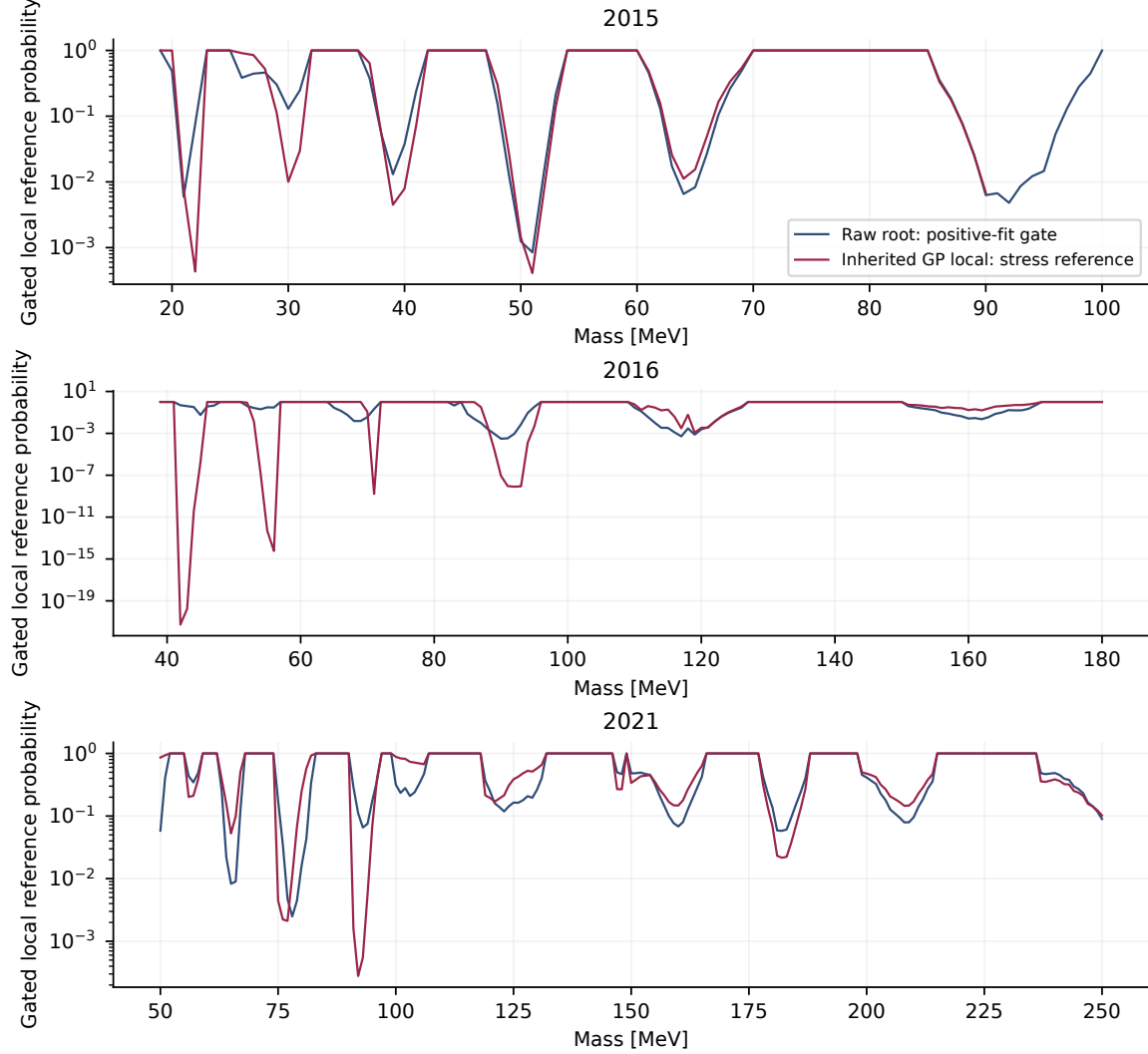
3. Fixed-mass local p-value curves



The one-positive-component reference is $p_0 = \Phi[-\max(0, r)]$, where r is the signed profile-likelihood root. The deficit convention gives $p_0 = 0.5$. These are fixed-mass asymptotic references, not a recalibration of the significance GP or a probability for a pair of resonances.

4. Relation to the new significance-GP regions

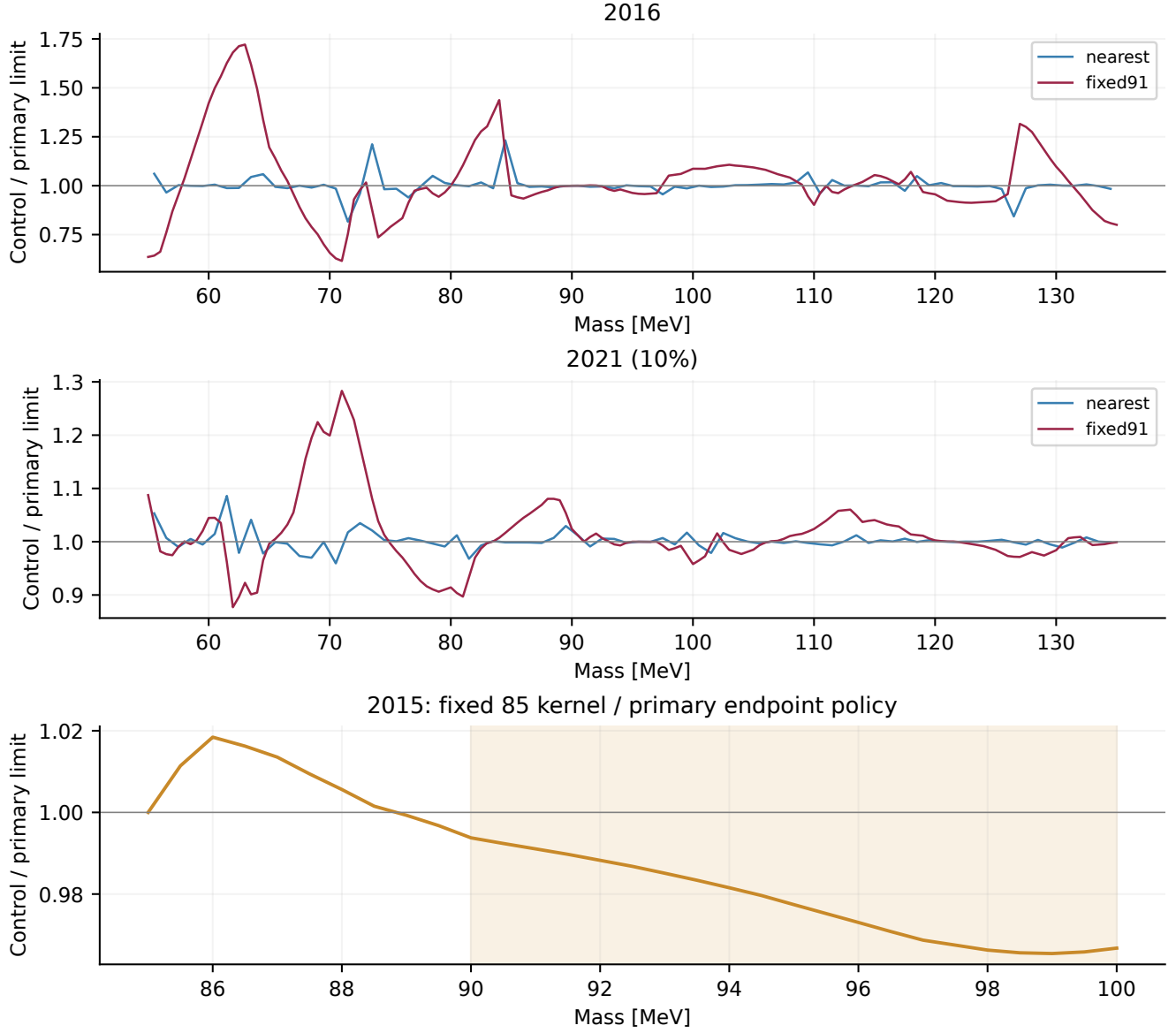
The previous GP probability field uses a stress-centered score, $(r_m - a_m)/s_m$, with the positive-fit gate $r_m > 0$. It can rank locations differently from the raw root because its reference mean is not zero. The earlier 2016 field was particularly sensitive to a negative stress offset. That field guides the observed regional catalogue; its single-mass trials factor is not transferred to a two-mass or shifted-mass search.



Pinned integer-grid GP local probabilities compared with the raw-root reference. Both plotted curves use probability one when the positive-fit gate fails, solely to show the same selection convention. The preceding page instead uses the conventional $p_0 = 0.5$ deficit value. No GP covariance is interpolated to manufacture a newly calibrated half-MeV probability field.

Moving mass hypotheses reuse counts and sidebands. In the parent 2021 field, correlations of the 71 MeV dip with the 65 and 78 MeV endpoints are -0.75 and -0.65 ; conditioning on the endpoints leaves a dip residual of only -0.52 conditional standard deviations. This explains why peak and trough probabilities cannot be multiplied. The new pair likelihood includes the intervening bins once, with its own profiled background covariance [1].

5. Kernel-state and endpoint controls



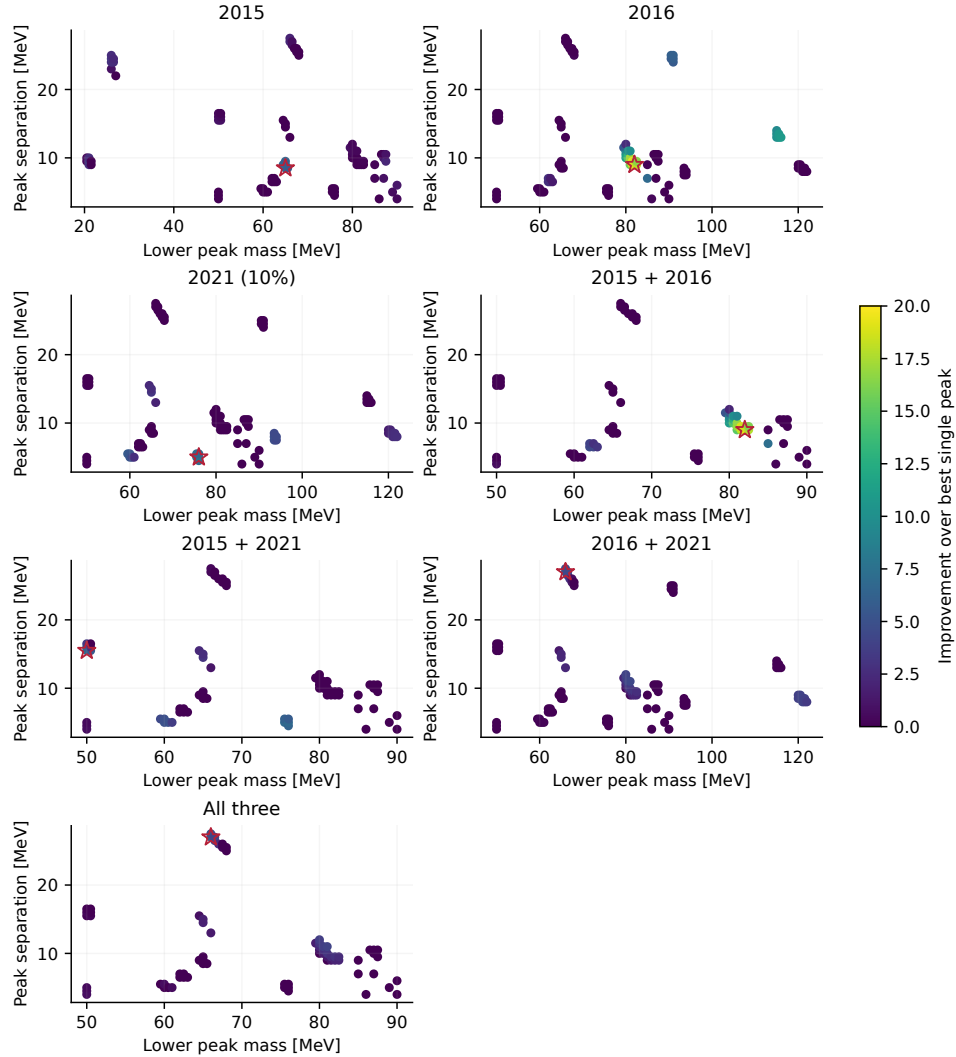
Ratios to the primary half-MeV upper limits. Nearest means the closest archived integer kernel, with half-integer ties rounded upward. Fixed91 holds the 91 MeV state over the whole high-mass diagnostic interval. The final panel compares 2015 with a frozen 85 MeV state; above 90 the primary uses the 90 MeV state. These test background-policy dependence and are not alternative curves selected for favorable results.

Across these individual half-integer controls, the largest nearest-state limit change is 23.1%. All original integer kernel knots are unchanged; the 2016 signed-root replay agrees within 3.3×10^{-11} .

6. Targeted two-component catalogue

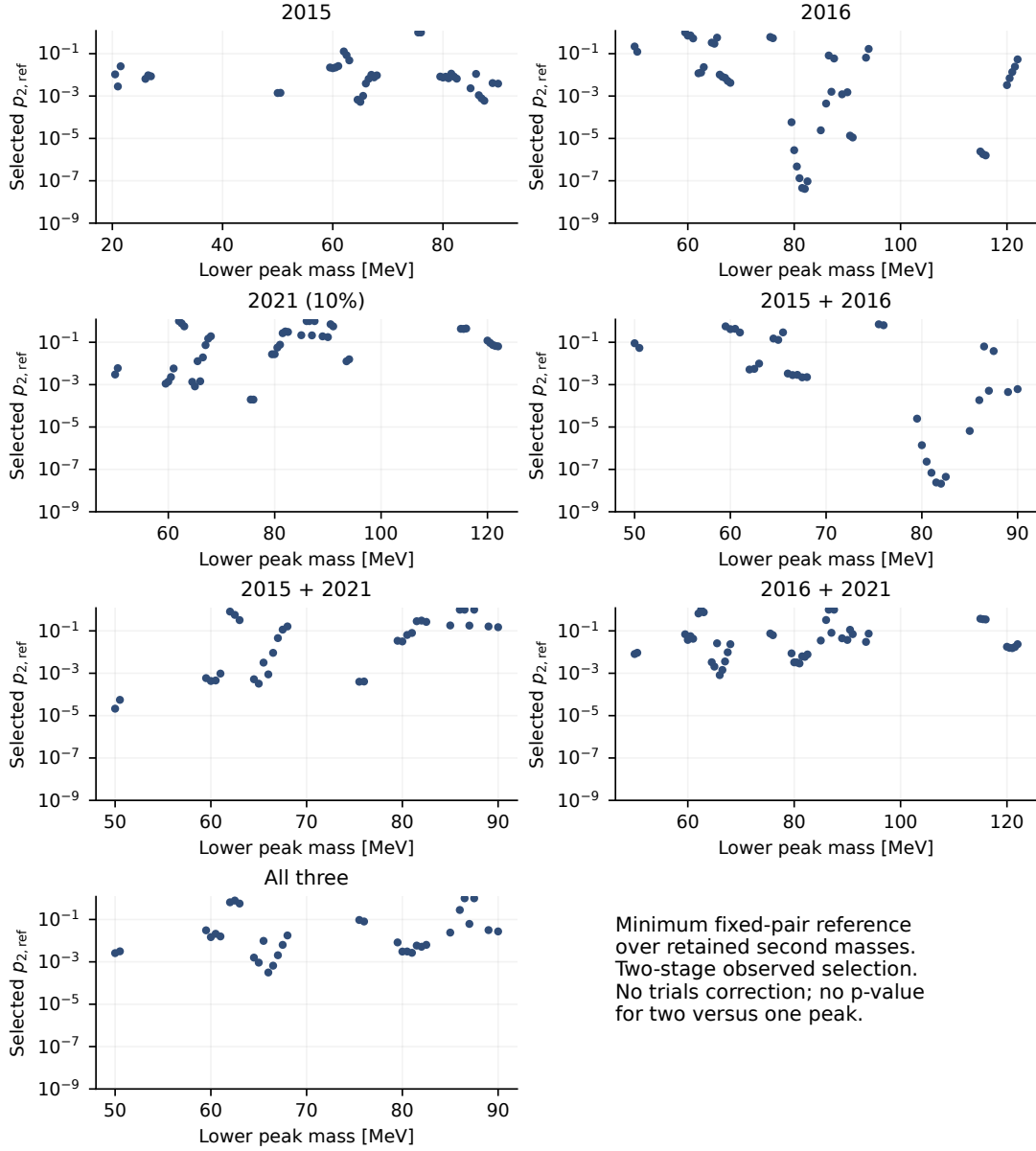
Requested regions are interpreted as lower-peak bands of ± 5 MeV around the displayed centers. Parent regional leaders seed half-MeV stencils. A coarse extension tests second masses through 100 MeV in 2015-containing scopes. All retained pair coordinates are evaluated in every eligible search. Screening by the endpoint improvement reduces 2,968 scope fits to 600 fully recomputed comparisons at 112 pair coordinates. Every requested regional view is represented. This bounded two-stage selection does not prove a maximum over all pairs.

For every pair, $\Omega_d = [m_1 - 3\sigma_d(m_1), m_2 + 3\sigma_d(m_2)]$ is removed from training. All hypotheses use identical count bins and the same GP constraint, with the archived kernel at the floored midpoint. The positive strengths are shared between campaigns. We compare the pair with a nonnegative single peak scanned across the full fixed region, clipped to the new scope domain, and locally refine its best integer mass.



Color is $\Delta_{2/1} = \max[0, 2(\ell_{1,\text{best}} - \ell_2)]$. Signed differences remain in the tables. Stars identify each new catalogue leader. Mass selection and scope selection are explicit; these colors are not significance contours.

7. Two-component reference p-values



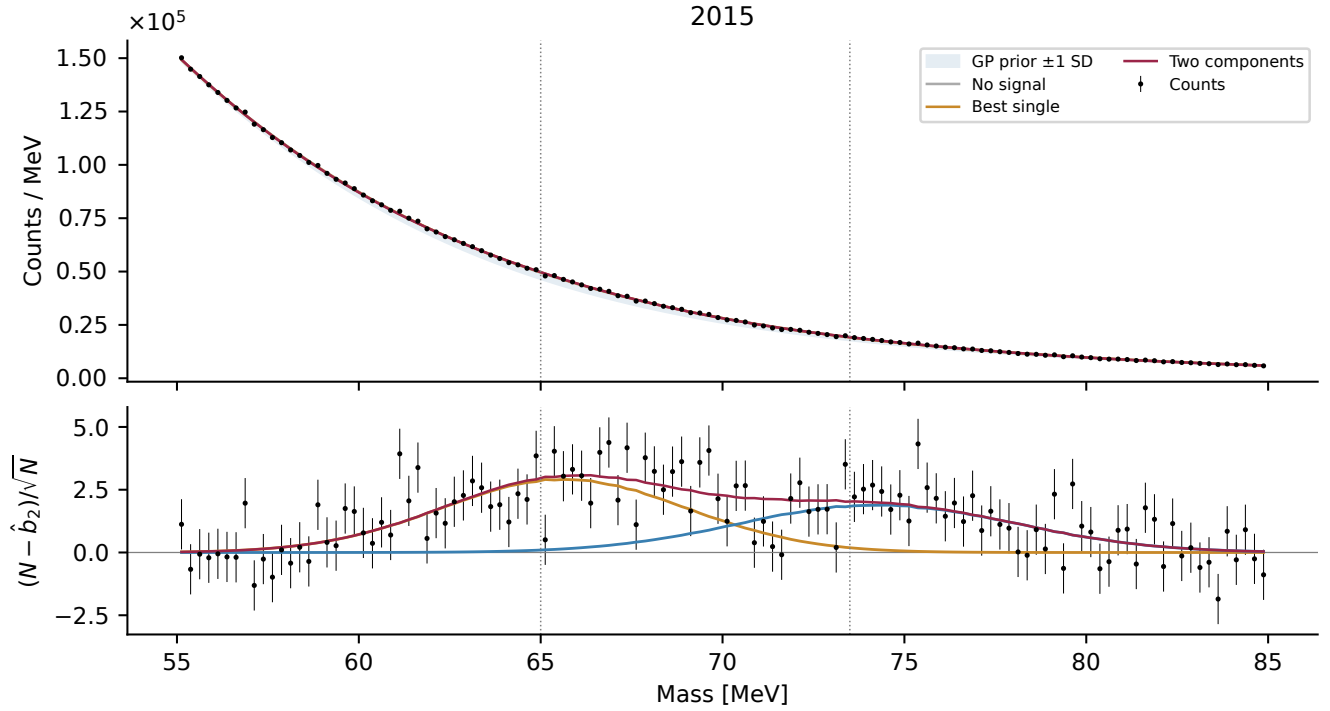
For each retained lower mass, points show the smallest fixed-pair $Q_2 = 2(\ell_0 - \ell_2)$ reference probability over retained upper masses. Large Q_2 can arise from one component alone. Selecting the pair invalidates interpreting these values as global probabilities.

The fixed-pair quadratic reference is the two-boundary chi-bar mixture with weights $w_1 = 1/2$, $w_2 = \arccos(c)/(2\pi)$, and $w_0 = 1/2 - w_2$, where c is the efficient null-score correlation after profiling the GP nuisances. For positive Q_2 , $p_{2,\text{ref}} = \frac{1}{2} \Pr(\chi_1^2 \geq Q_2) + w_2 \Pr(\chi_2^2 \geq Q_2)$. This correlation differs from the moving-mask field correlation. No p-value is assigned to the selected, nonnested pair-versus-best-single comparison.

8. Regional results: 2015

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 20$	21 / 30.5	9.96	3.17	2.84×10^{-3}
$m_1 \simeq 30$	26 / 50.5	8.26	3.17	6.46×10^{-3}
$m_1 \simeq 65$	65 / 73.5	13.68	6.88	5.32×10^{-4}
2015 extension	66 / 93	9.06	2.29	3.91×10^{-3}

The displayed scope leader is 65/73.5 MeV; its best single mass is 64.768 MeV. The fitted-amplitude correlation is 0.792.



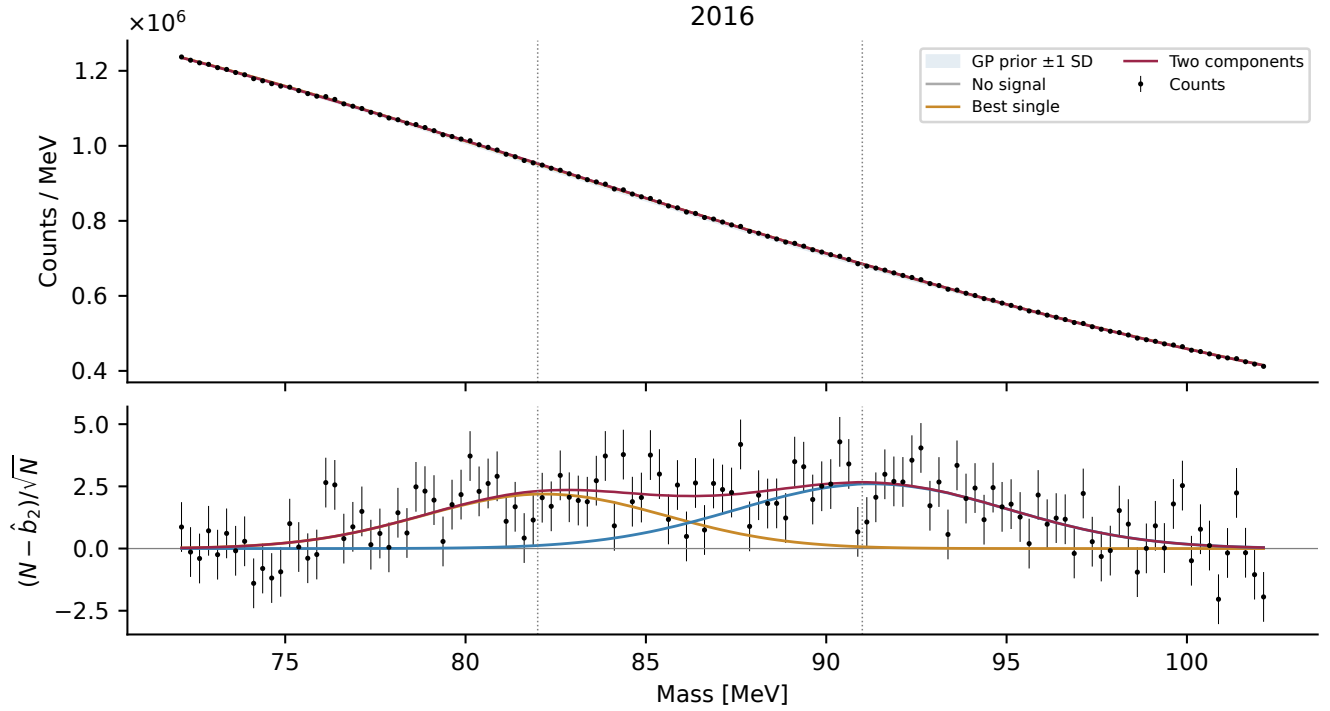
The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

The table lists one leader per regional view, so nearby views can select the same pair. The complete catalogue retains additional separated candidates. Further regional extraction displays appear in the appendix; all numerical fits and surfaces are supplied digitally.

9. Regional results: 2016

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 65$	62 / 69	7.53	2.79	1.19×10^{-2}
$m_1 \simeq 80$	82 / 91	32.32	19.57	4.15×10^{-8}
$m_1 \simeq 90$	85 / 92	19.45	6.83	2.41×10^{-5}
$m_1 \simeq 120$	116 / 129	25.08	10.81	1.58×10^{-6}

The displayed scope leader is 82/91 MeV; its best single mass is 90.394 MeV. The fitted-amplitude correlation is 0.662.



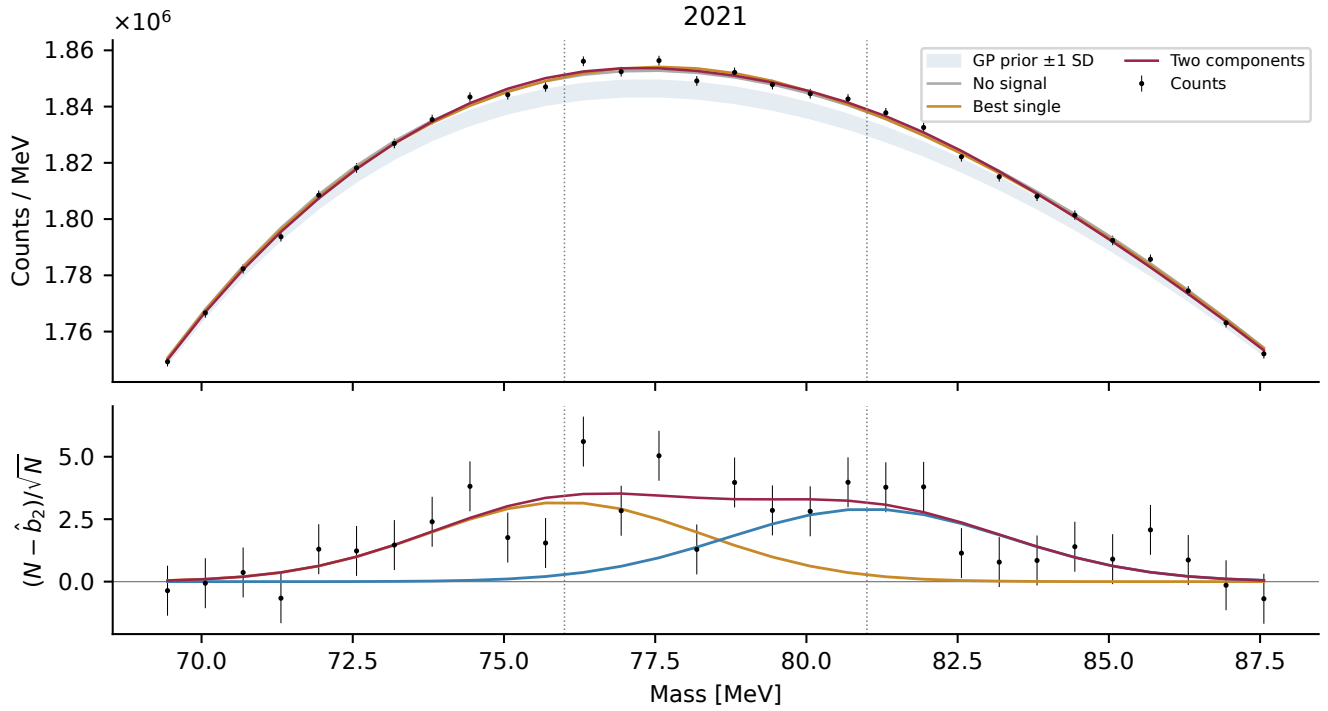
The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

The table lists one leader per regional view, so nearby views can select the same pair. The complete catalogue retains additional separated candidates. Further regional extraction displays appear in the appendix; all numerical fits and surfaces are supplied digitally.

10. Regional results: 2021 (10%)

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 65$	60 / 65	11.77	5.34	1.39×10^{-3}
$m_1 \simeq 80$	76 / 81	15.43	7.71	1.97×10^{-4}
$m_1 \simeq 90$	93.5 / 101.5	7.33	4.84	1.27×10^{-2}
$m_1 \simeq 120$	122 / 130	4.16	2.72	6.42×10^{-2}

The displayed scope leader is 76/81 MeV; its best single mass is 78.344 MeV. The fitted-amplitude correlation is 0.567.



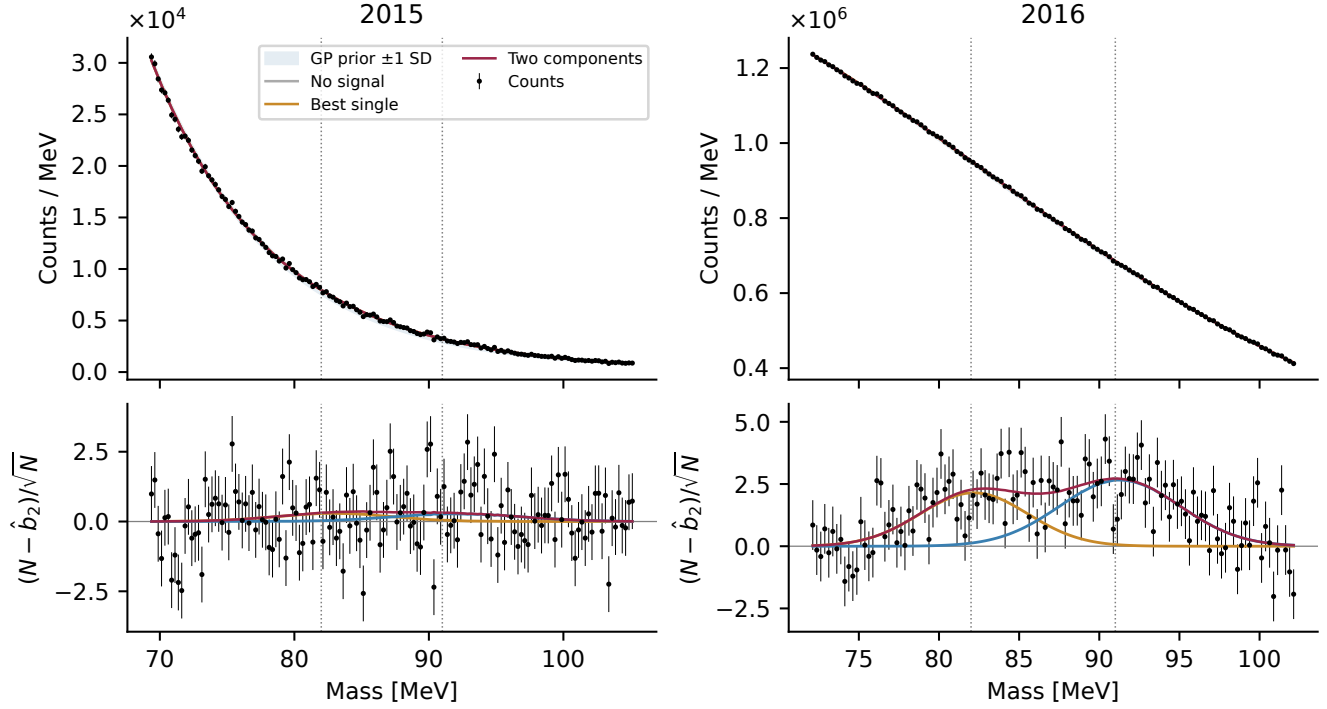
The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

The table lists one leader per regional view, so nearby views can select the same pair. The complete catalogue retains additional separated candidates. Further regional extraction displays appear in the appendix; all numerical fits and surfaces are supplied digitally.

11. Regional results: 2015 + 2016

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 65$	62 / 68.5	9.12	4.40	5.18×10^{-3}
$m_1 \simeq 80$	82 / 91	33.66	19.09	2.12×10^{-8}

The displayed scope leader is 82/91 MeV; its best single mass is 90.443 MeV. The fitted-amplitude correlation is 0.662.



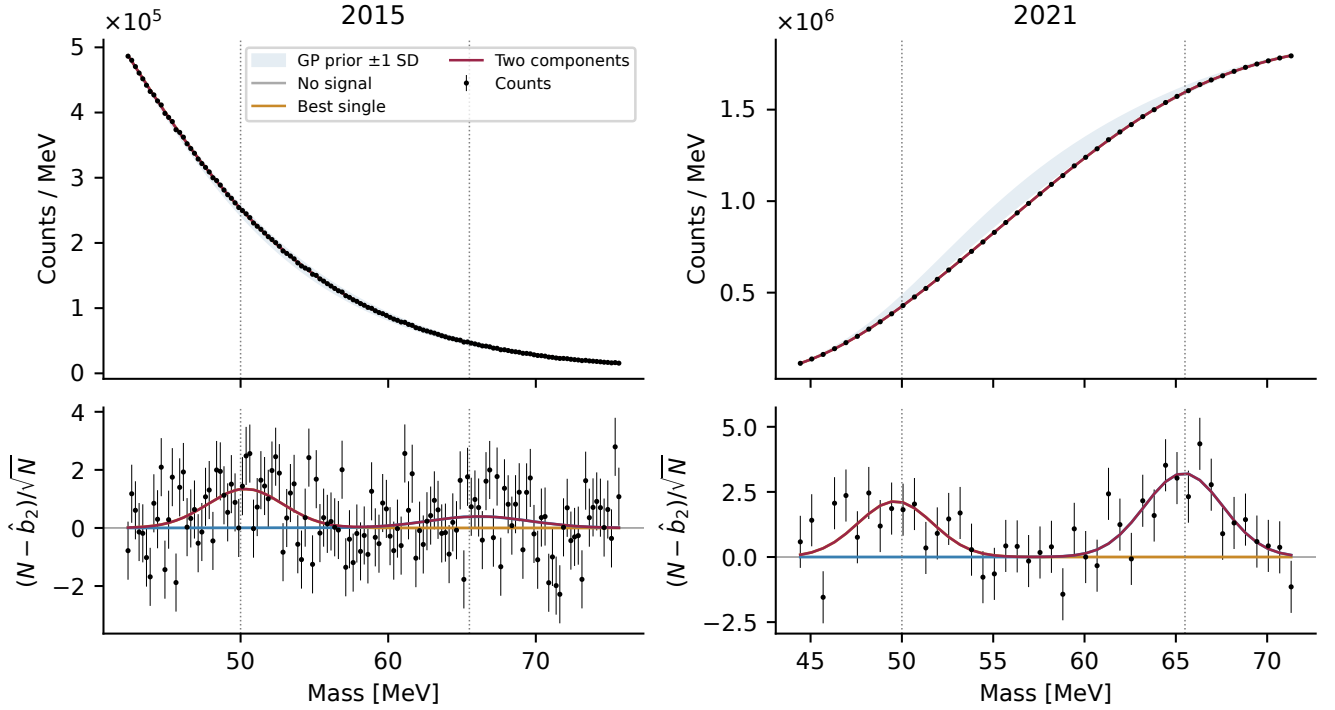
The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

The table lists one leader per regional view, so nearby views can select the same pair. The complete catalogue retains additional separated candidates. Further regional extraction displays appear in the appendix; all numerical fits and surfaces are supplied digitally.

12. Regional results: 2015 + 2021

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 50$	50 / 65.5	19.41	7.03	2.13×10^{-5}
$m_1 \simeq 62.5$	60.5 / 65.5	13.93	5.06	4.57×10^{-4}
$m_1 \simeq 75$	75.5 / 80.5	14.04	6.76	3.98×10^{-4}

The displayed scope leader is 50/65.5 MeV; its best single mass is 65.471 MeV. The fitted-amplitude correlation is 0.082.



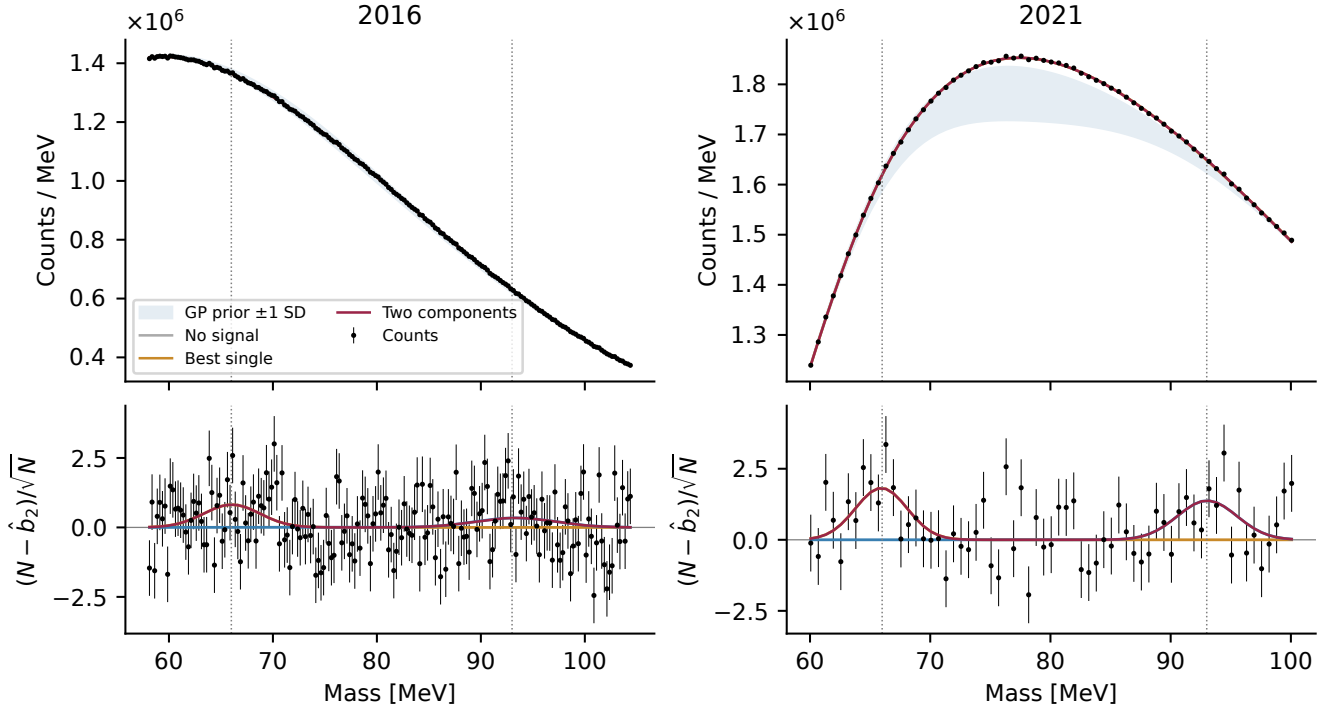
The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

The table lists one leader per regional view, so nearby views can select the same pair. The complete catalogue retains additional separated candidates. Further regional extraction displays appear in the appendix; all numerical fits and surfaces are supplied digitally.

13. Regional results: 2016 + 2021

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 65$	66 / 93	12.24	5.22	8.17×10^{-4}
$m_1 \simeq 80$	81 / 92	9.87	4.37	2.91×10^{-3}
$m_1 \simeq 120$	121 / 129.5	6.90	4.09	1.54×10^{-2}

The displayed scope leader is 66/93 MeV; its best single mass is 65.706 MeV. The fitted-amplitude correlation is 0.093.



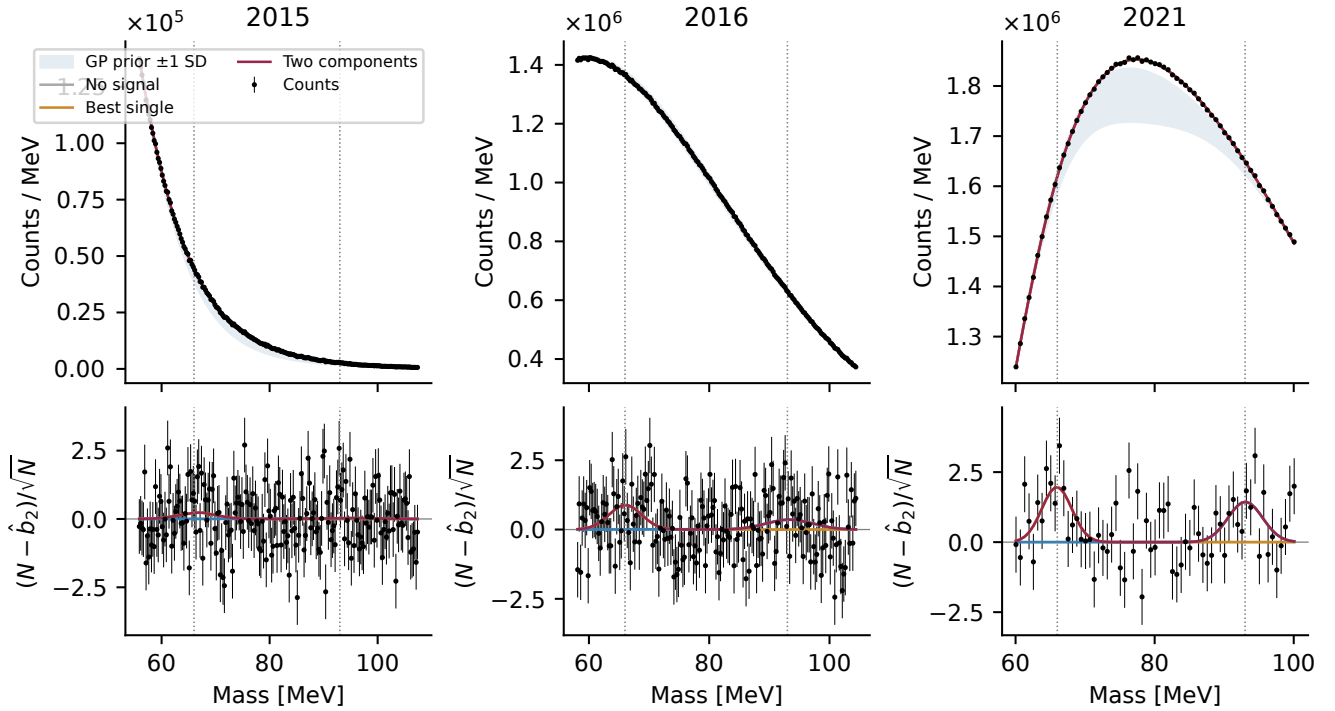
The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

The table lists one leader per regional view, so nearby views can select the same pair. The complete catalogue retains additional separated candidates. Further regional extraction displays appear in the appendix; all numerical fits and surfaces are supplied digitally.

14. Regional results: All three

Requested region	m_1/m_2 (MeV)	Q_2	$\Delta_{2/1}$	$p_{2,\text{ref}}$
$m_1 \simeq 50$	50 / 54	2.01	0.78	2.04×10^{-1}
$m_1 \simeq 62.5$	66 / 93	14.11	5.47	3.15×10^{-4}
$m_1 \simeq 65$	66 / 93	14.11	5.47	3.15×10^{-4}
$m_1 \simeq 75$	80 / 92	9.66	3.62	3.06×10^{-3}
$m_1 \simeq 80$	81 / 92	10.04	3.99	2.67×10^{-3}
$m_1 \simeq 90$	85 / 92	6.06	0.04	2.41×10^{-2}

The displayed scope leader is 66/93 MeV; its best single mass is 65.568 MeV. The fitted-amplitude correlation is 0.092.



The common-region count fit and fitted-background-subtracted display. The band is the diagonal GP prior uncertainty. Residual error bars show count errors only; the fitted background is uncertain and correlated, so these are not independent significance pulls. The two colored components and their sum use the same likelihood as the upper panel.

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15. The new 66/93 MeV union candidate

Within the expanded targeted catalogue, the maximal-available search includes all three campaigns at 66/93 MeV and gives $\Delta_{2/1} = 5.47$. The prior domain ended 2015 at 90 MeV and therefore could not evaluate this same all-three hypothesis. The apparent change of leader is a change in the declared candidate/domain search, not a newly calibrated global preference.

Padding	$\Delta_{2/1}$ observed	D_{rate} observed	q_{add} stress
2.25	5.64	11.16	0.00
3	5.47	10.42	0.00
4	5.07	16.09	0.00

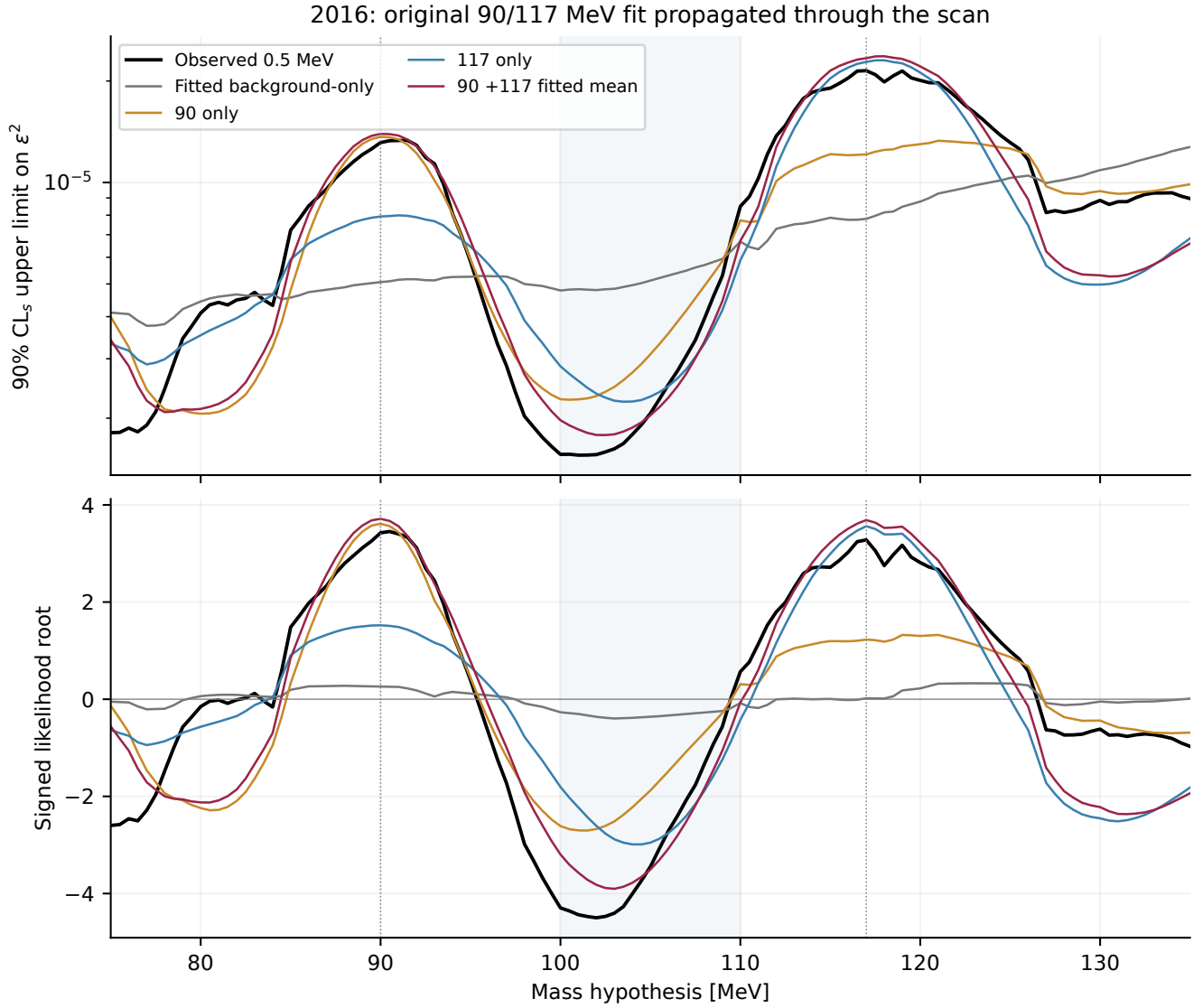
The pair-versus-single improvement remains about 5.1–5.6 over the three window choices. The rate-comparison deviance is 10.42 for the baseline window, comparing common strengths with independently fitted strengths on the same data and background constraints. This quantity is descriptive; the masses were selected and amplitudes can lie on boundaries.

The archived stress continuum gives $Q_2 \simeq 87$ but zero endpoint improvement: only the lower component is positive. Its best shifted single peak is slightly better than the fixed pair. This is a direct example of why a tiny fixed-pair null probability does not imply two positive signals.

Campaign	N_1 shared	N_2 shared	N_1 independent	N_2 independent
2015	828	43	3308	1120
2016	13975	5433	8639	13390
2021	16959	14481	19220	10042

These are full-support model yields using the retained conversion, not independently counted event samples. The selected pair is worth keeping in the catalogue, but the rate and background comparisons remain part of its interpretation.

16. Section 21 revisited with actual fine upper limits



Original 2016 90/117 MeV common-region fits propagated through the moving-mask analysis. Every colored curve starts from a full expected count spectrum and recomputes GP predictions and CL_s roots at every mass. The blue interval is the fixed 100–110 MeV trough window. These are deterministic post-fit predictions, not additional measurements.

The old two-component prediction tracked signed roots near 102–103 MeV closely, but overpredicted the negative response around 130 MeV. The actual limit calculation preserves the central trough while exposing its mass offset. Across 55–135 MeV, the log-limit RMS difference from data is 0.296 for the two-component mean, versus 0.629 for the fitted background-only mean. The same comparison gives correlation 0.928 versus 0.623. These shape metrics use correlated mass points and have no chi-square interpretation.

Positive components can generate a negative moving-mask response between them: when the trough is tested, the peaks can enter the GP training sidebands and raise the interpolated background. In these fitted reconstructions, the trough is part of the estimator response and cannot be counted as an independent third piece of particle evidence.

17. Two- and three-component injection experiment

The pair is at 90/117 MeV; the triple is at 65/90/117 MeV. Both use one common, broad fitted continuum per campaign, with the entire three-component interval excluded from training and the kernel anchored at 91 MeV. The 2016 region is 57.01–131.97 MeV, and the 2021 region is 58.63–125.59 MeV. The 2015 extension remains an observed-data study; it does not support a 117 MeV signal hypothesis.

The nonnegative three-component fit gives zero strength at 65 MeV in 2016 and at 117 MeV in 2021. To test three genuinely positive injected components, the reference strength is explicitly defined as

$$A_{dj}^{\text{ref}} = \max(\hat{A}_{dj}^{H_{123}}, \sigma_{dj}^{\text{Fisher}}).$$

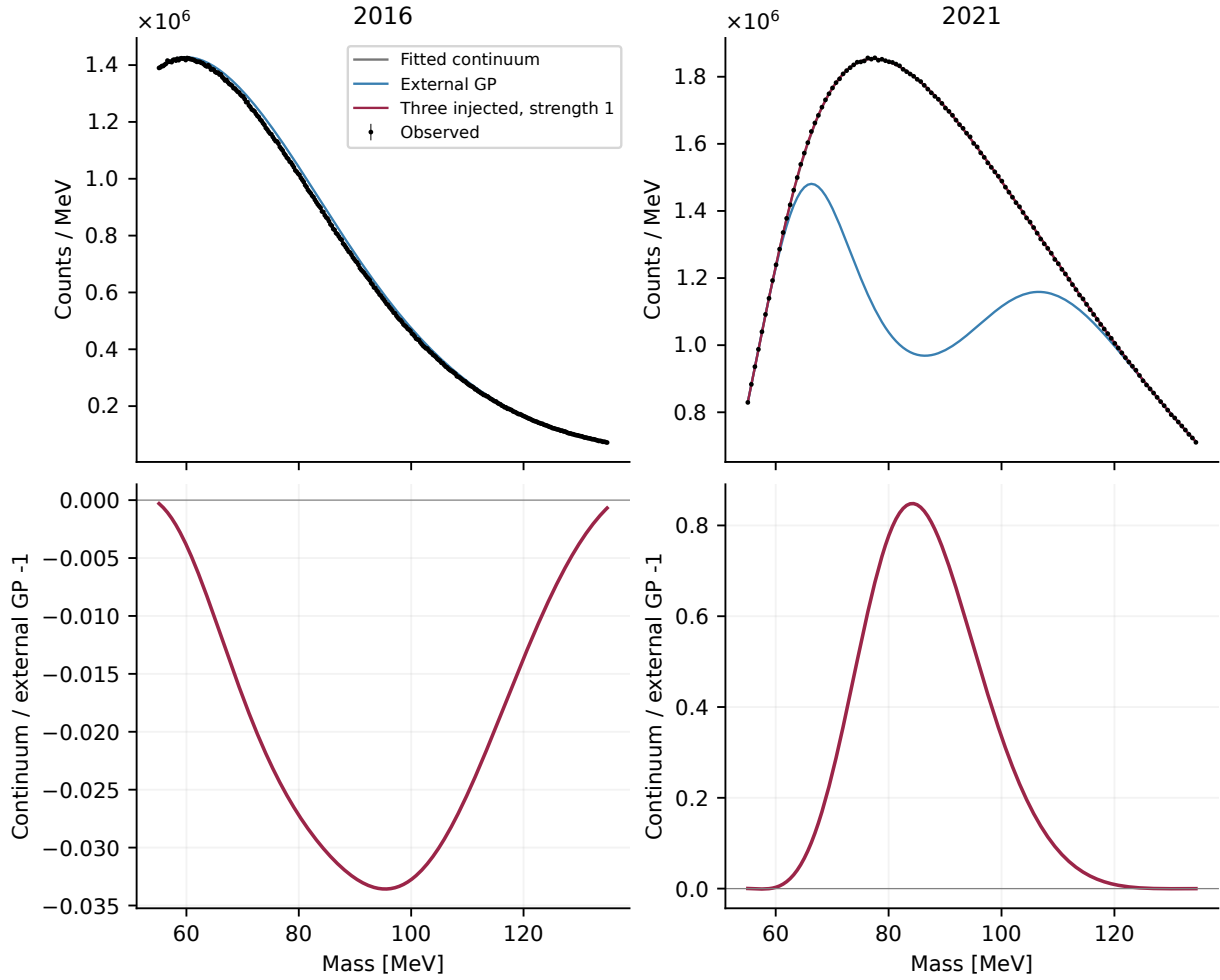
The Fisher scale uses the efficient expected information at the fitted same-region null. It is a benchmark floor, not a measured signal. The untouched fitted means are also propagated separately.

Campaign	Mass	$\hat{\epsilon}^2 [10^{-6}]$	$\epsilon_{\text{ref}}^2 [10^{-6}]$	Reference yield
2016	65	0.000	2.638	7069
2016	90	5.924	5.924	11463
2016	117	9.869	9.869	6909
2021	65	6.311	6.311	19667
2021	90	0.580	1.777	8411
2021	117	0.000	1.605	6226

The table expresses the coupling-like strengths as ϵ^2 in units of 10^{-6} . At strength multiplier a , the full-spectrum truth is the common fitted continuum plus a times the two or three indicated reference components. The pair omits only the 65 MeV component. Campaign amplitudes are independent in this morphology generator; the shared-rate combined extraction can consequently be misspecified. A separate common-rate fitted mean tests that physical assumption deterministically.

Multipliers are 0, 0.5, 1, 1.5, and 2. There are exactly ten complete joint Poisson experiments per scheme and multiplier: 100 labels, with the identical zero-strength schemes reusing ten baselines, hence 90 distinct joint spectra. Independent background draws and independent half-strength signal increments are added cumulatively. This gives correct Poisson marginals while pairing comparisons. Independent campaigns remain independent within each experiment.

18. What the fitted continuum contributes



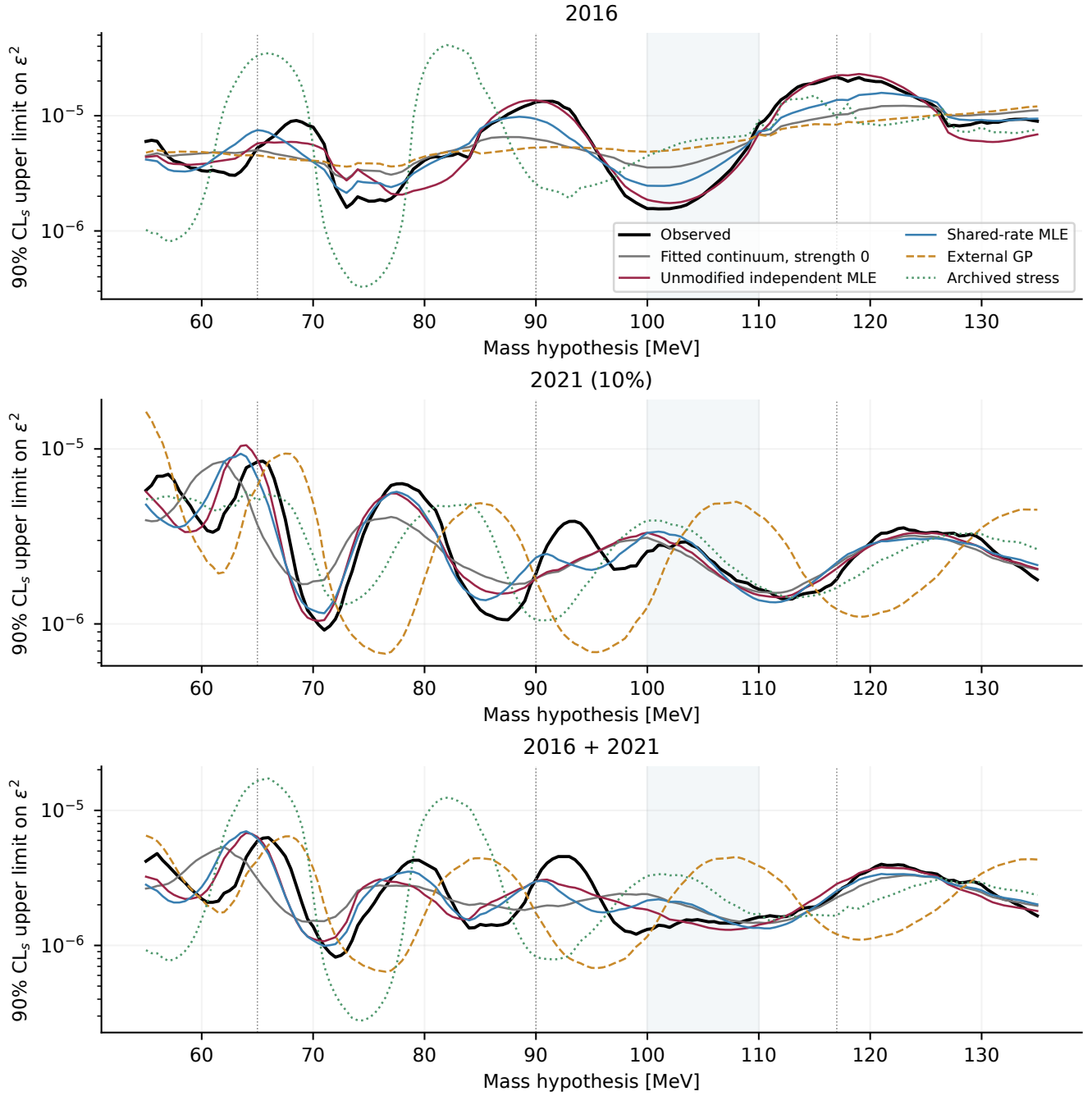
The broad-region fitted continuum, unshifted external-GP mean, and positive three-component reference spectrum. The lower panels expose the fitted background displacement rather than hiding it inside a signal prediction. Analysis-bin counts are retained.

The fitted 2021 continuum differs from the external GP mean by as much as 84.81% inside the broad omitted region, near 84.44 MeV. The largest 2016 difference is 3.36%. Positivity of the full count mean does not validate a broad interpolation: this large 2021 displacement makes the baseline choice a material part of any successful reconstruction.

The pair and triple use exactly the same continuum, so their difference isolates the added 65 MeV benchmark component within this construction. The zero-strength curve still contains the continuum fitted under the three-component hypothesis; it is not an independently established no-signal truth. The external GP and archived stress controls provide different deterministic baselines, with their own disclosed limitations.

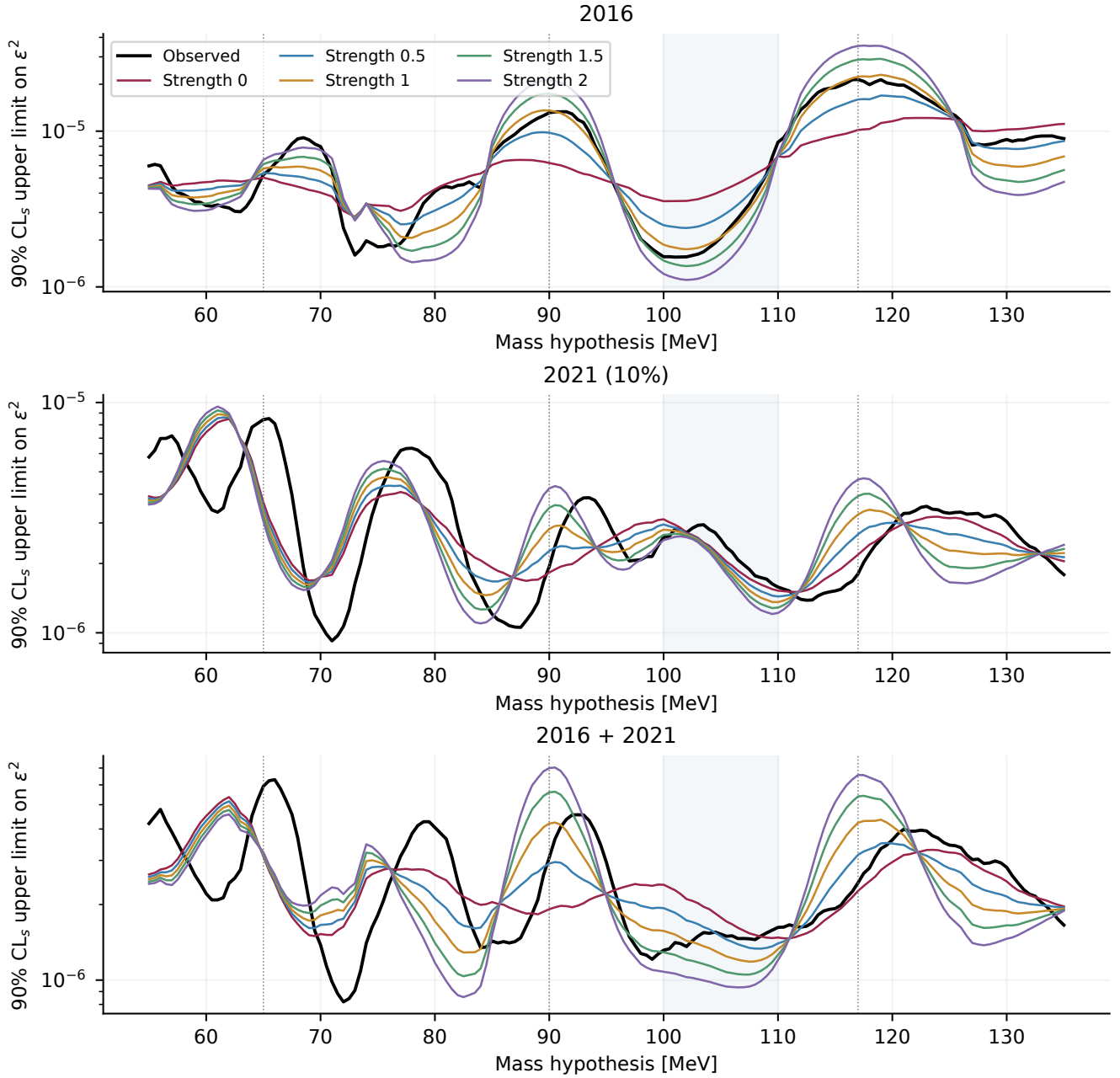
Each pseudoexperiment regenerates full-support counts, then refits the GP prediction and covariance from its own sidebands at every moving mask. Kernel coordinates, resolution and density-to-yield conversion remain frozen. No per-mass independent count draws, random Gaussian score fields, or extra toys are used.

19. Deterministic background and rate controls



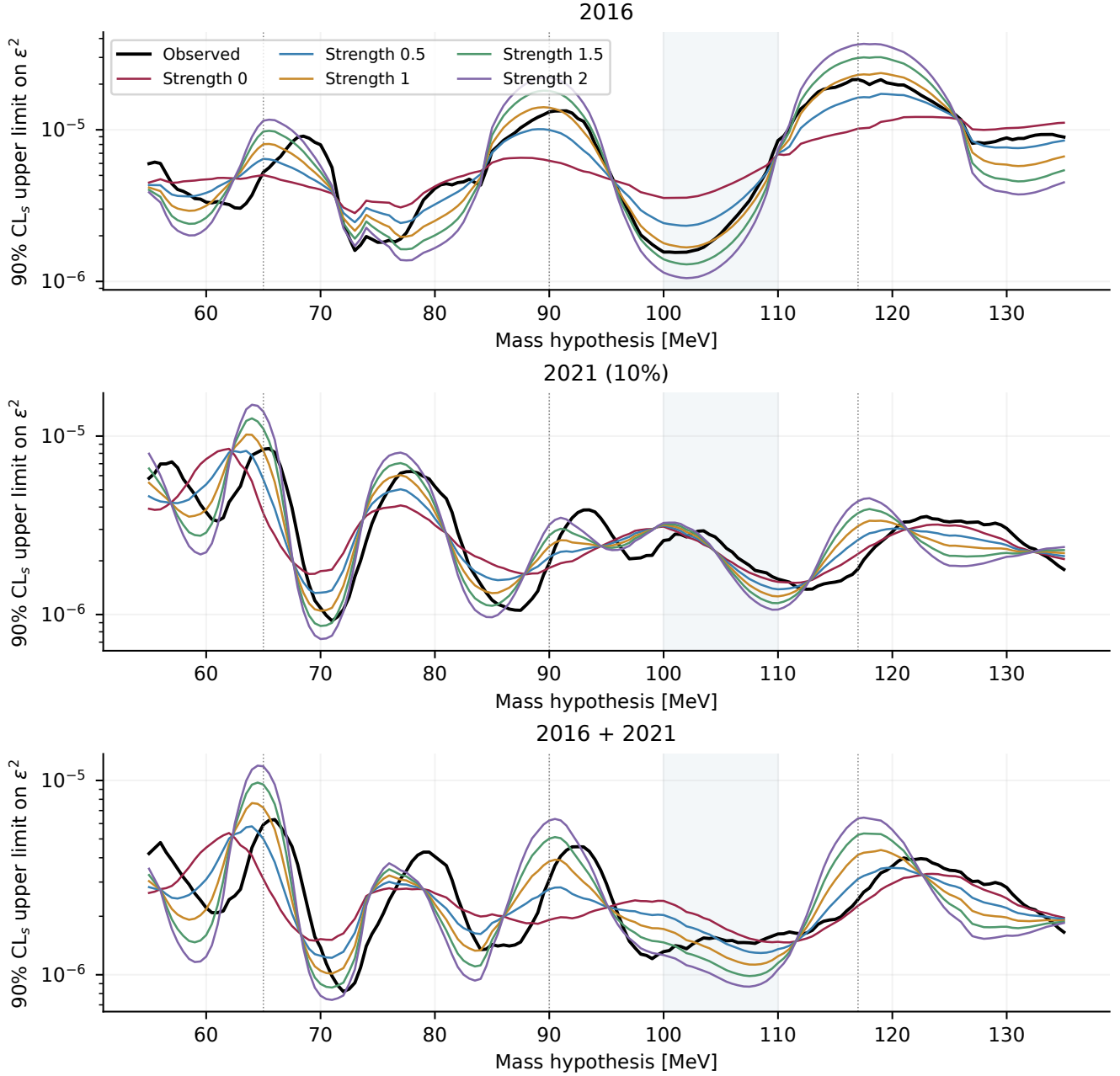
Actual upper-limit scans of the untouched independent fitted mean, shared-rate fitted mean, fitted continuum alone, external-GP mean, and archived stress mean. All use the same mass window and upper-limit calculation. A close post-fit curve reuses the data and is not an independent goodness-of-fit test.

20. Unfluctuated two-component strength scan



Full expected count spectra at all five strength multipliers, propagated through GP fitting and bounded CL_s extraction. The black curve is observed. Vertical lines mark 65, 90 and 117 MeV; only the triple injects at 65 MeV. The fixed trough interval and outer tails are included in every comparison.

21. Unfluctuated three-component strength scan

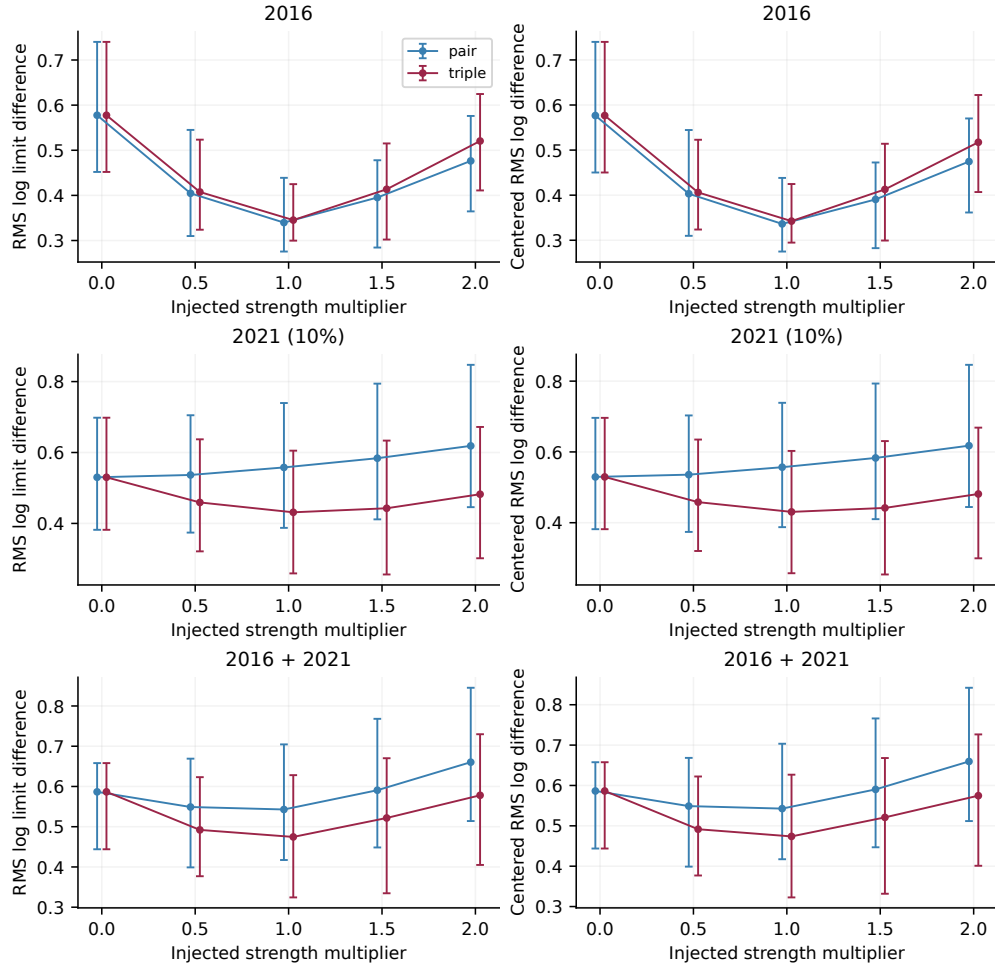


Full expected count spectra at all five strength multipliers, propagated through GP fitting and bounded CL_s extraction. The black curve is observed. Vertical lines mark 65, 90 and 117 MeV; only the triple injects at 65 MeV. The fixed trough interval and outer tails are included in every comparison.

22. What the ten-trial pilot reproduces

For each complete curve, let $u_m = \log[U_m^{\text{trial}}/U_m^{\text{observed}}]$. We record $D = [\langle u_m^2 \rangle]^{1/2}$ across 55–135 MeV and the centered version $D_c = [\langle (u_m - \langle u \rangle)^2 \rangle]^{1/2}$. The latter removes an overall multiplicative offset. These uniformly weighted descriptive distances are not sums of independent residuals.

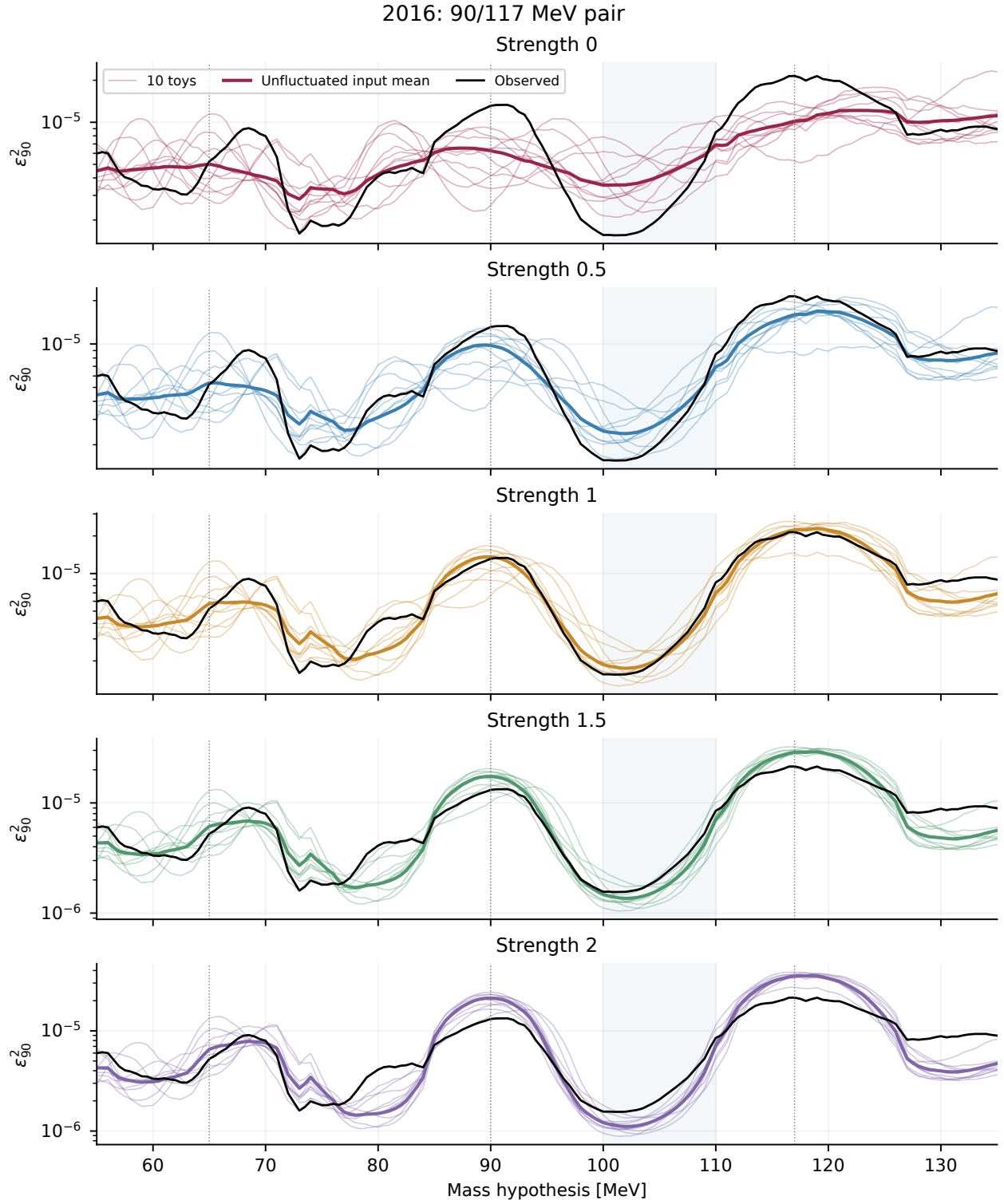
Search	Baseline D	Pair, $a = 1$	Triple, $a = 1$
2016	0.578	0.340	0.345
2021 (10%)	0.530	0.558	0.431
2016 + 2021	0.587	0.543	0.475



Medians and observed minimum-to-maximum ranges among the ten paired trials. These ranges are not confidence bands. Strength one is the best of the tested multipliers by median full-window distance for the two-component 2016 and three-component 2021/combined comparisons. This pilot ranking is itself descriptive.

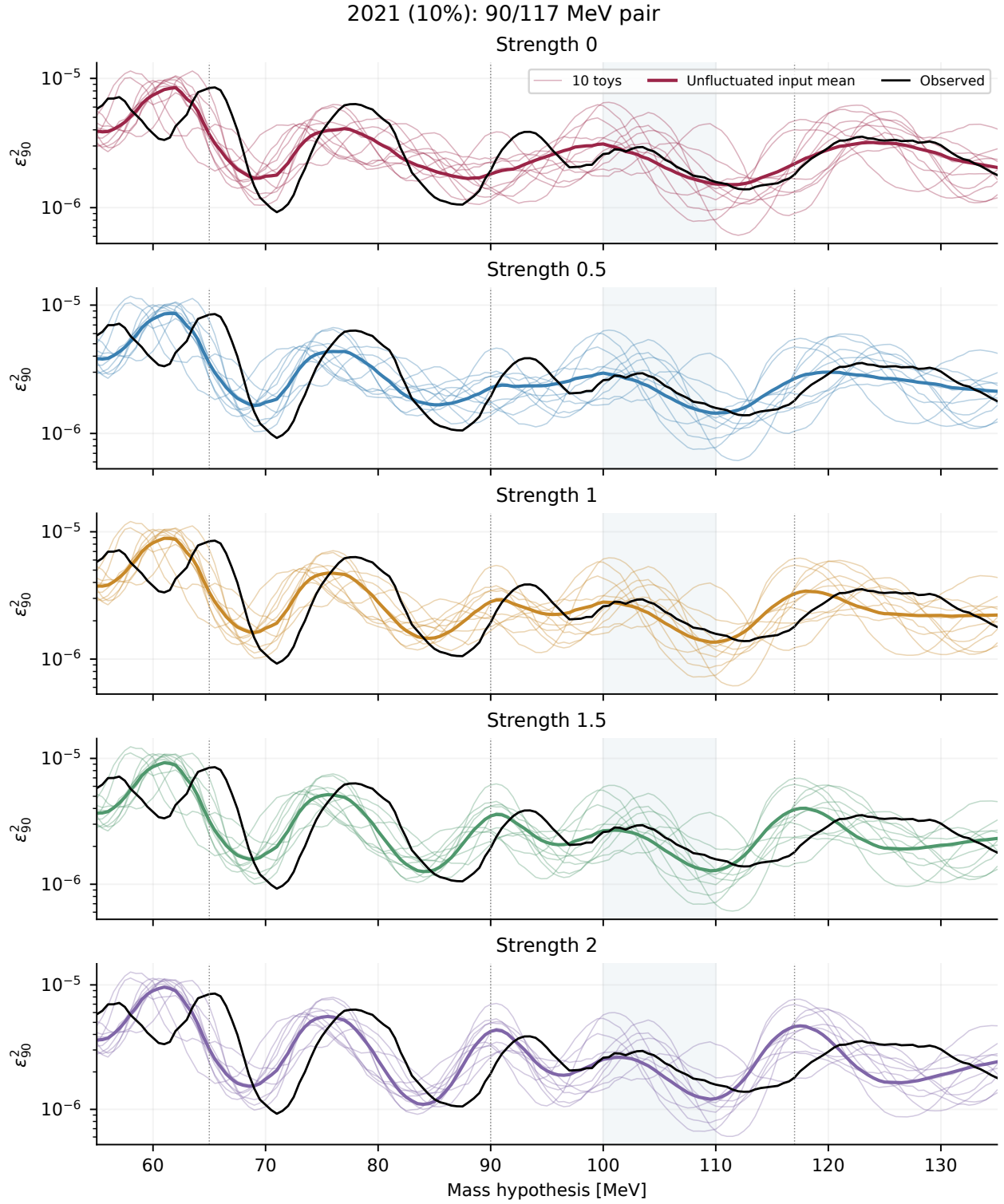
Stronger injections can deepen the trough while worsening the full-window match. For the 2016 pair, the number of trials with at least the observed fixed-window trough contrast rises from 0/10 at strength 0 to 5/10 at strength 1 and 10/10 at strength 2. Those fractions describe the chosen post-fit generator; they are not background p-values. The following pages show every trial rather than a fitted uncertainty band.

23. Ten-trial curves: 2016, 90/117 MeV pair



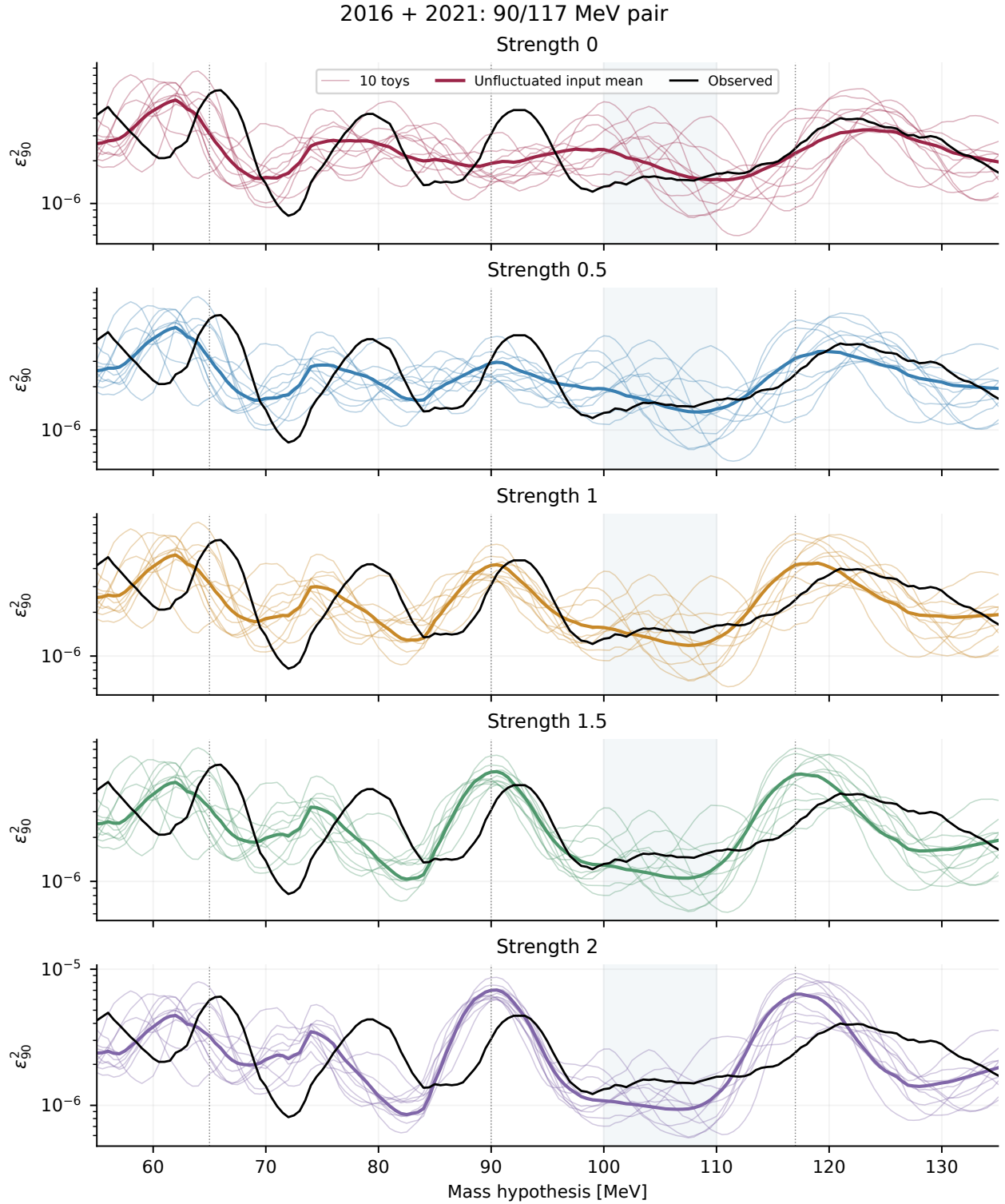
All ten complete-experiment curves at each strength. Thin colored curves are trials, the thick colored curve is the unfluctuated input mean propagated through the estimator, and black is observed. The zero-strength experiments are shared between the two schemes. Moving masses and overlapping searches reuse each experiment, so the plotted curves are correlated.

24. Ten-trial curves: 2021 (10%), 90/117 MeV pair



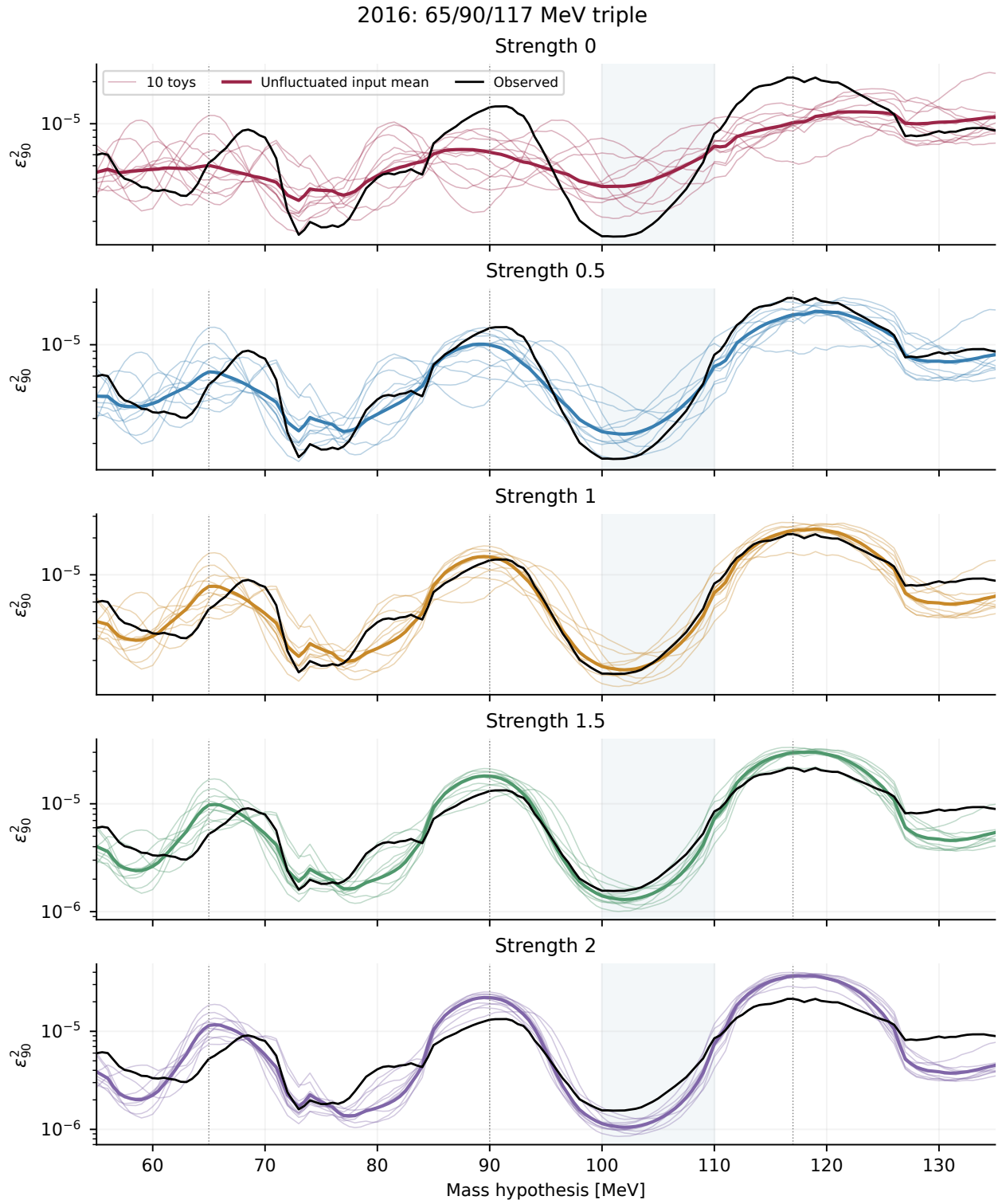
All ten complete-experiment curves at each strength. Thin colored curves are trials, the thick colored curve is the unfluctuated input mean propagated through the estimator, and black is observed. The zero-strength experiments are shared between the two schemes. Moving masses and overlapping searches reuse each experiment, so the plotted curves are correlated.

25. Ten-trial curves: 2016 + 2021, 90/117 MeV pair



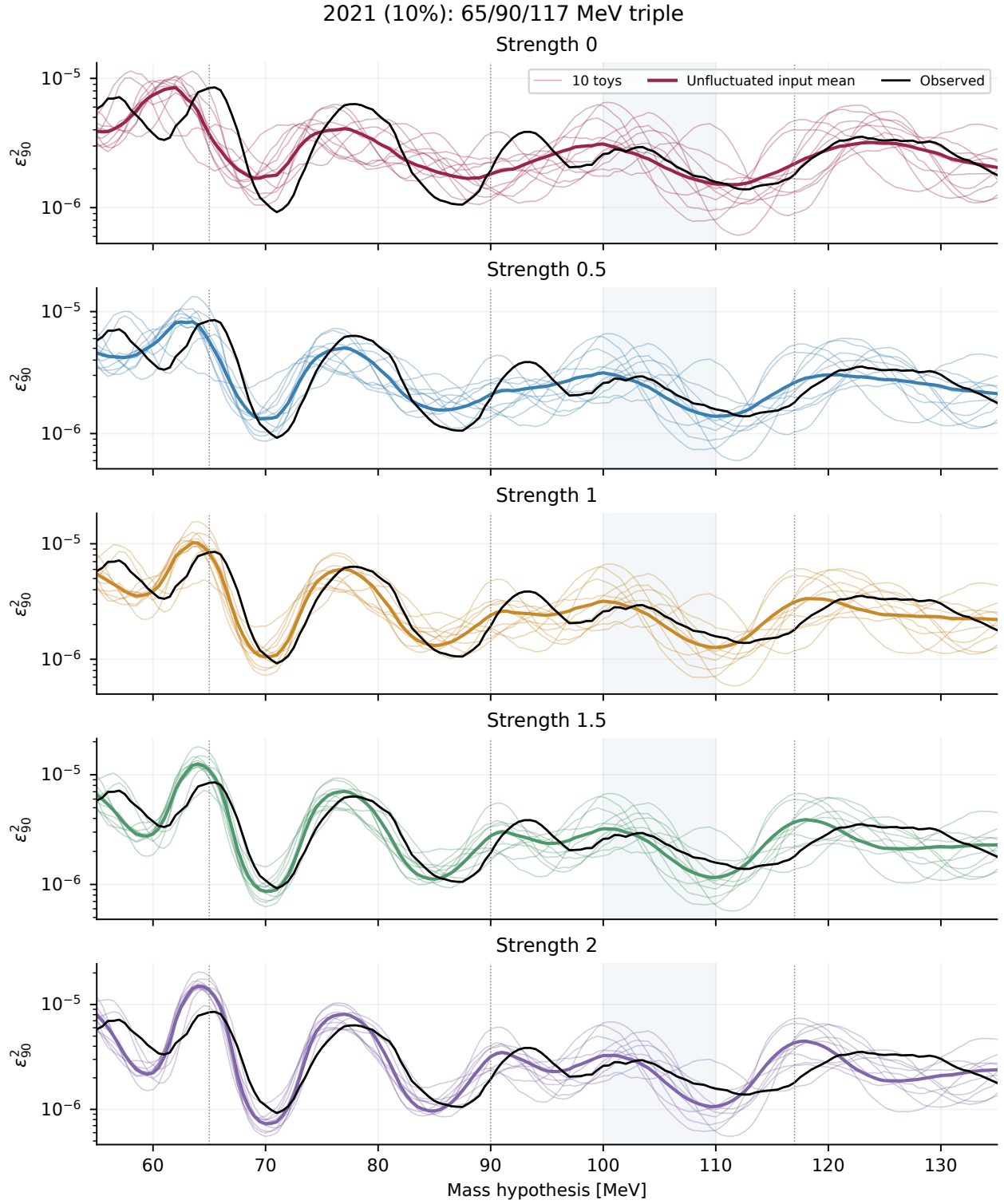
All ten complete-experiment curves at each strength. Thin colored curves are trials, the thick colored curve is the unfluctuated input mean propagated through the estimator, and black is observed. The zero-strength experiments are shared between the two schemes. Moving masses and overlapping searches reuse each experiment, so the plotted curves are correlated.

26. Ten-trial curves: 2016, 65/90/117 MeV triple



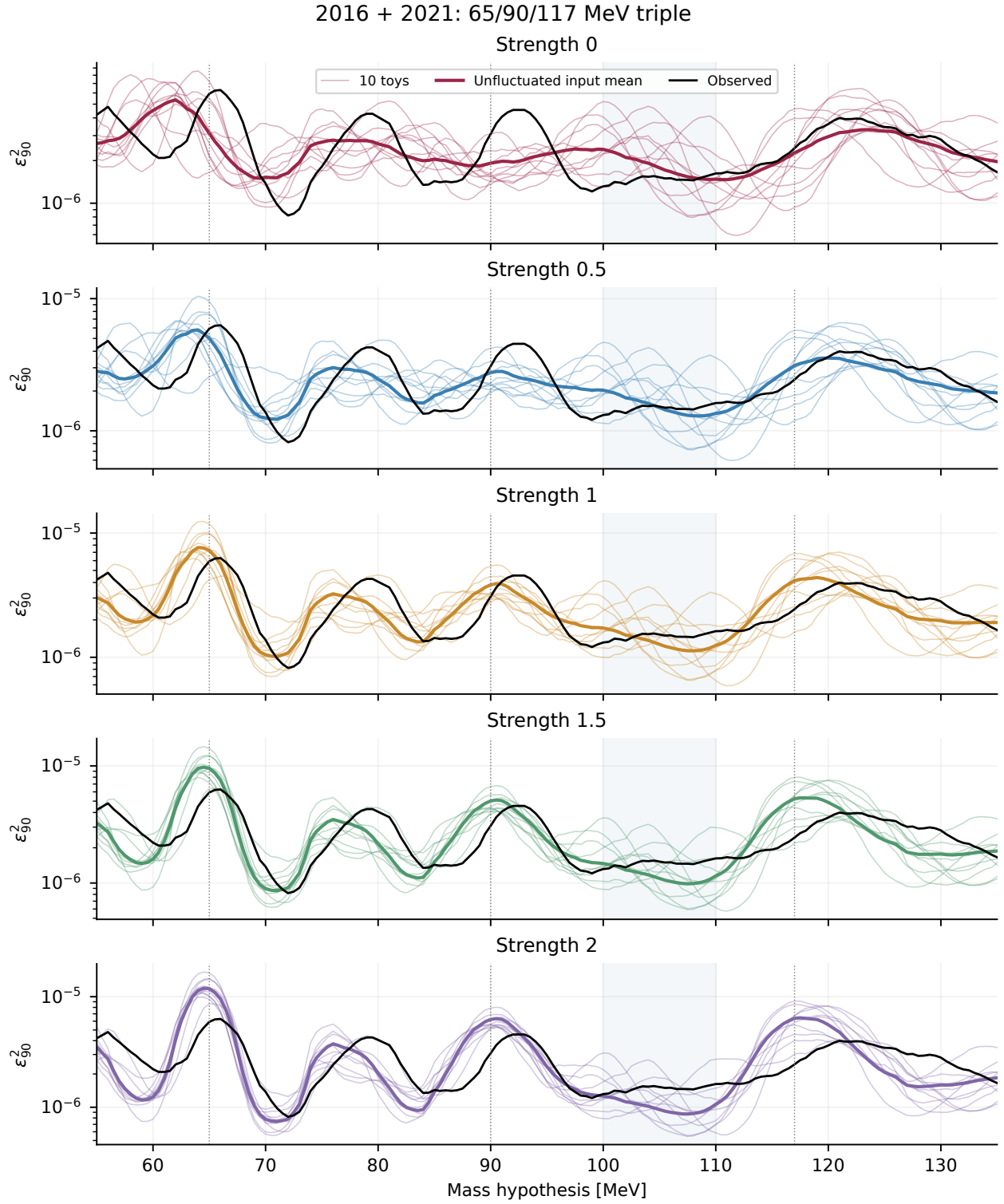
All ten complete-experiment curves at each strength. Thin colored curves are trials, the thick colored curve is the unfluctuated input mean propagated through the estimator, and black is observed. The zero-strength experiments are shared between the two schemes. Moving masses and overlapping searches reuse each experiment, so the plotted curves are correlated.

27. Ten-trial curves: 2021 (10%), 65/90/117 MeV triple



All ten complete-experiment curves at each strength. Thin colored curves are trials, the thick colored curve is the unfluctuated input mean propagated through the estimator, and black is observed. The zero-strength experiments are shared between the two schemes. Moving masses and overlapping searches reuse each experiment, so the plotted curves are correlated.

28. Ten-trial curves: 2016 + 2021, 65/90/117 MeV triple



All ten complete-experiment curves at each strength. Thin colored curves are trials, the thick colored curve is the unfluctuated input mean propagated through the estimator, and black is observed. The zero-strength experiments are shared between the two schemes. Moving masses and overlapping searches reuse each experiment, so the plotted curves are correlated.

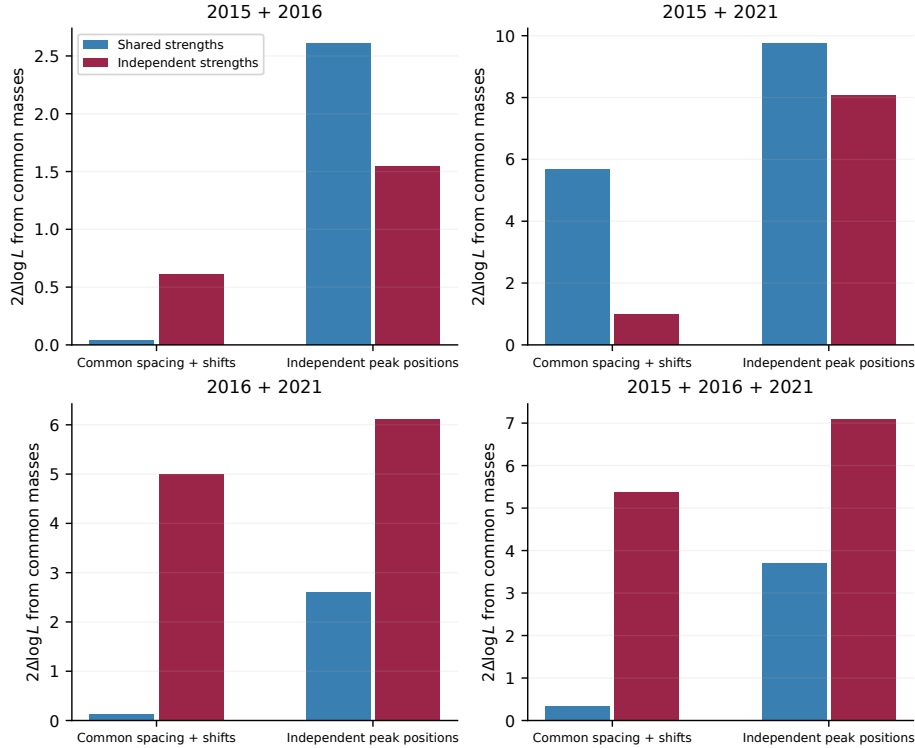
29. Shared separation with year-dependent shifts

To test a common shape without requiring identical reconstructed masses, the two peak locations are written as

$$m_{1d} = C + \delta_d - \Delta/2, \quad m_{2d} = C + \delta_d + \Delta/2.$$

In the common-spacing model the first campaign fixes $\delta = 0$, and other offsets lie in $[-3, 3]$ MeV. We compare common positions, common spacing with offsets, and independent pairs. The independent-pair model expands non-reference center domains by 3 MeV; its relative offsets need not obey the 3 MeV bound. Every comparison within a family retains one fixed union region covering all allowed masses and their 3σ tails, one fixed kernel anchor, and identical count bins. Shared and independently fitted strengths are separate branches.

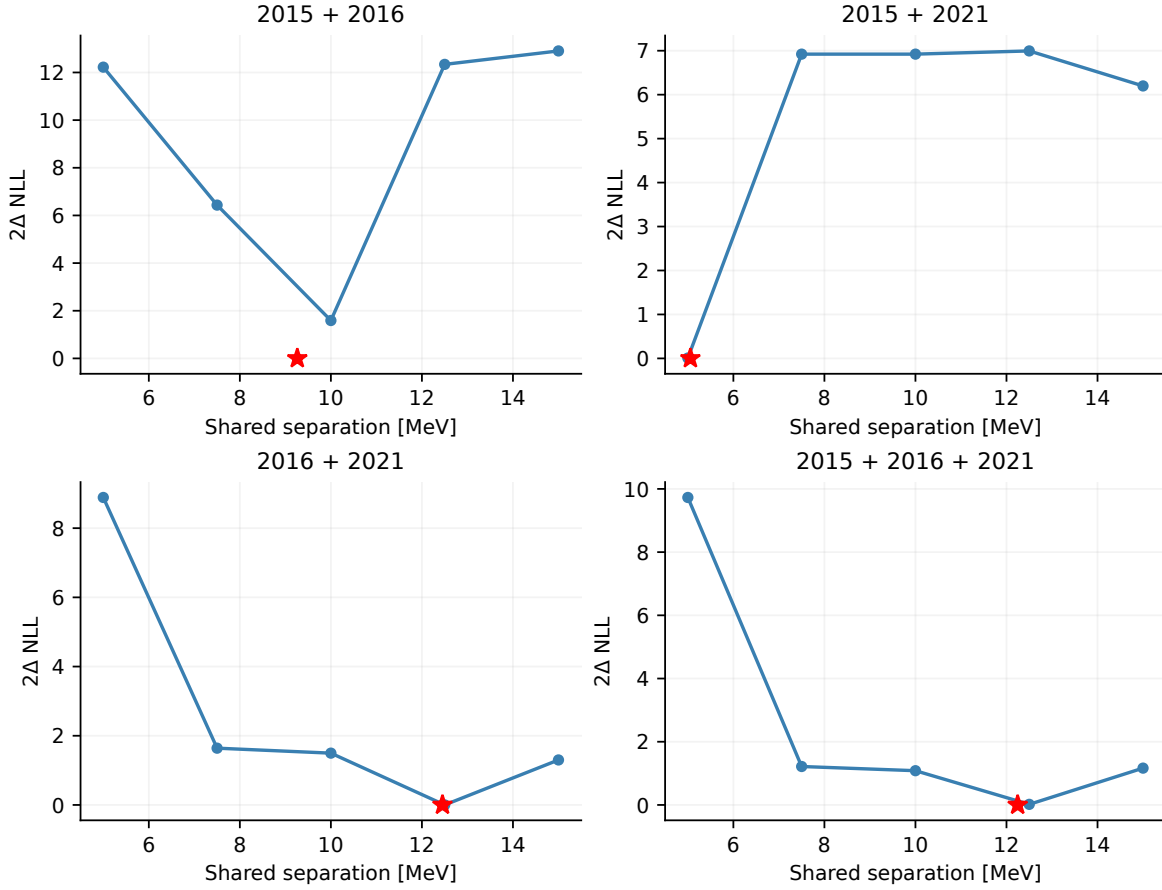
Family	C domain (MeV)	Δ domain (MeV)	Searches
80 MeV family	80–87	5–15	Four combinations
Lower family	70–76	10–20	2016 + 2021
Wide family	100–106	22–32	2016 + 2021



Likelihood gains relative to common reconstructed masses, within each fixed experiment. Blue shares component strengths across campaigns; red fits them independently. No significance is assigned to these selected mass-model comparisons.

An additive reconstruction offset preserves the separation; a multiplicative momentum-scale error changes it as well. The allowed offsets here are exploratory bounds, not measured calibration errors or Gaussian calibration constraints. Beam energy can affect production rates, acceptance and resolution, but does not itself change a resonance's invariant mass. Connecting the fitted offsets to that mechanism would require event-level detector and calibration information.

30. Spacing profiles and rate compatibility

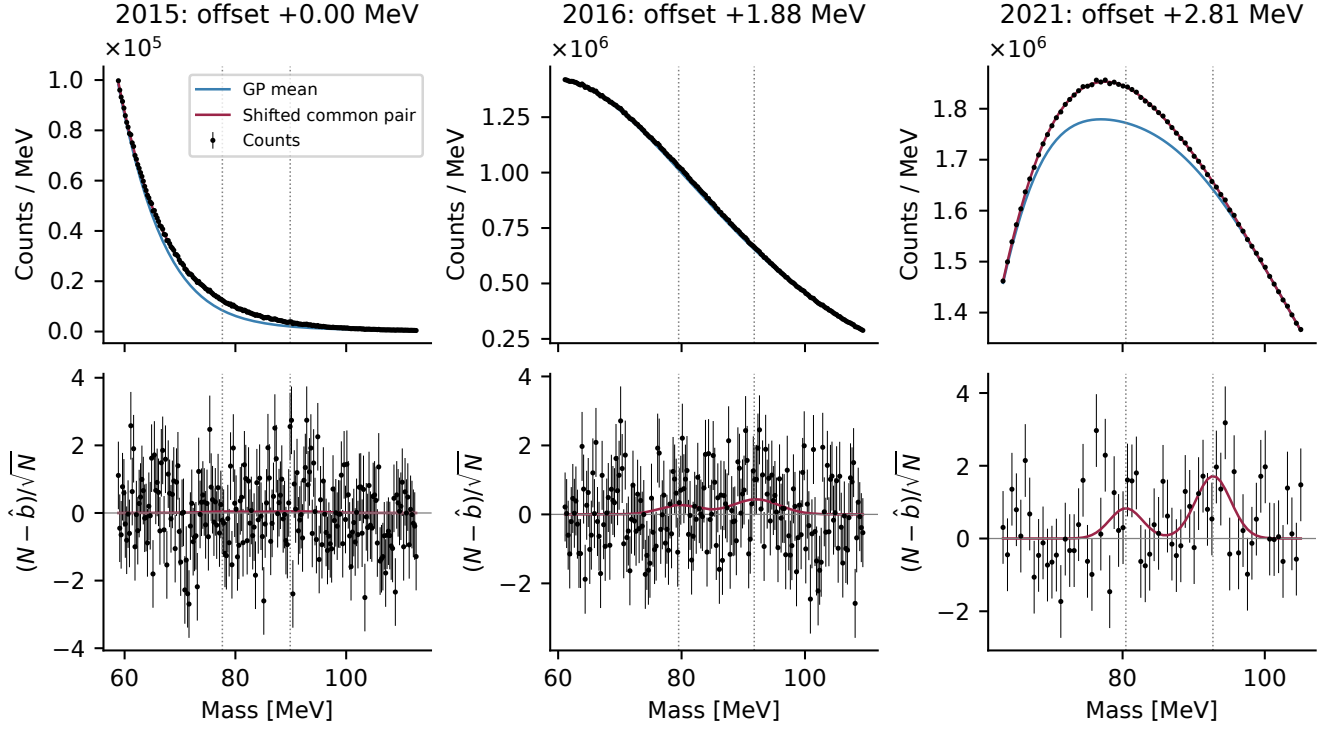


Coarse profiles at five declared separations, with the locally optimized shared-strength solution marked by a star. Centers and permitted relative offsets are reprofiled. These are likelihood shapes, not confidence intervals or a proof of a continuous global optimum.

Family / campaigns	Δ (MeV)	Shift gain	Rate gain	Bounds
80s: 2015+2016	9.26	0.04	21.54	0
80s: 2015+2021	5.06	5.67	22.09	1
80s: 2016+2021	12.45	0.13	15.69	0
80s: 2015+2016+2021	12.25	0.34	43.11	0
Lower: 2016+2021	14.10	2.46	0.25	0
Wide: 2016+2021	29.19	3.32	9.10	0

The all-three 80 MeV family changes only modestly when offsets are allowed, whereas independently fitted rates improve it substantially. In 2015+2021 the shared-strength offset reaches -3 MeV and the separation sits close to its 5 MeV lower bound. Across the 36 models, twelve touch a mass bound and seven contain a zero component. Their mass estimates must be read with those limitations.

31. The all-three common-spacing count fit

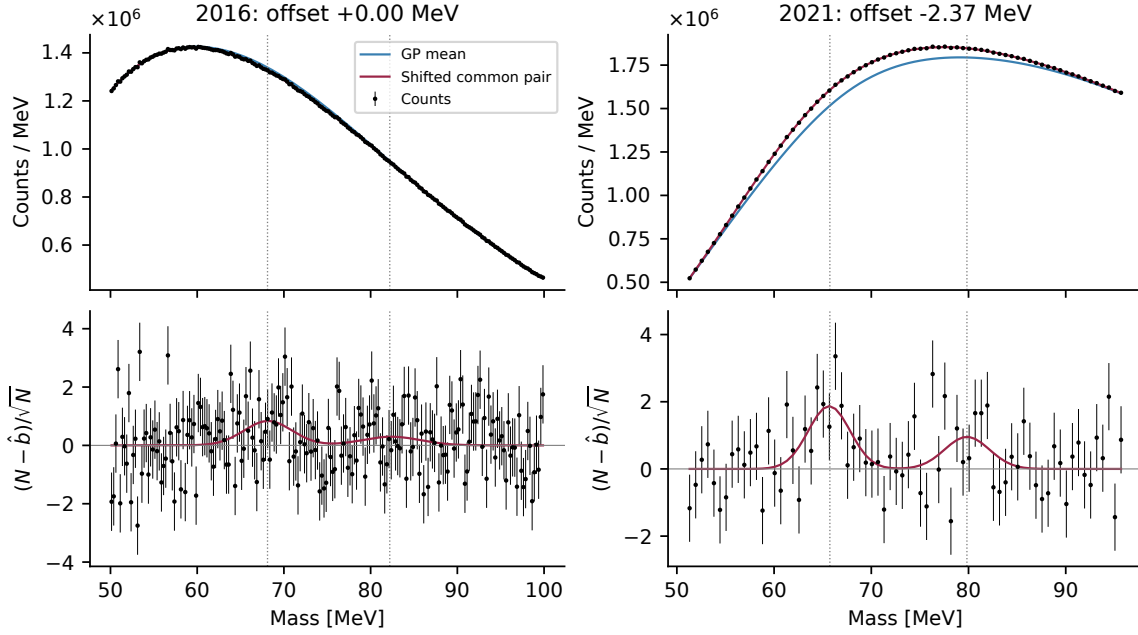


Shared strengths and shared separation, with individual reconstruction offsets. All allowed mass models use the same wider region and GP constraint. The residual display is for visualizing components and does not remove background uncertainty.

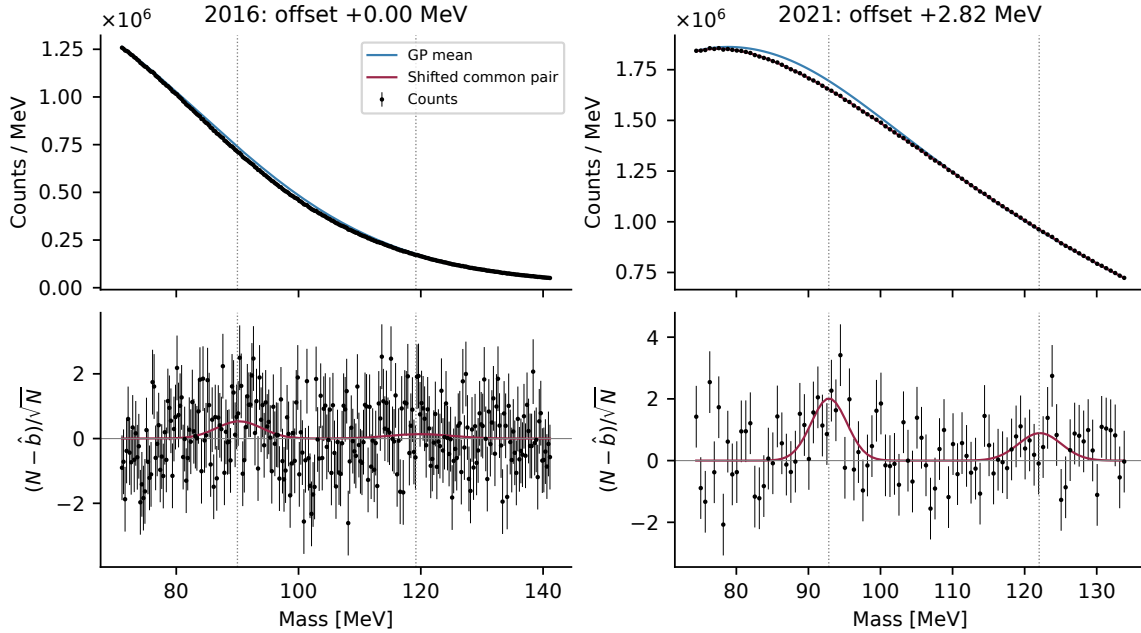
Campaign	m_1 (MeV)	m_2 (MeV)	δ (MeV)
2015	77.649	89.895	0.000
2016	79.527	91.773	1.878
2021	80.459	92.705	2.810

The common separation is 12.246 MeV. The same fit has a rate-comparison gain of 43.11 when strengths are released. Under independently fitted rates the preferred spacing becomes about 9.74 MeV and one campaign component reaches zero. The claimed similarity is therefore sensitive to how the signal rates are related, rather than a stable identification of the same two resonances in every campaign.

32. Lower and wider common-spacing alternatives



Lower family: $\Delta = 14.10$ MeV and a 2021 offset of -2.37 MeV. Independent peak positions improve the shared-strength fit by only 0.008 in twice log likelihood; releasing campaign rates gives 0.25. This is the clearest descriptive common-spacing example among these bounded alternatives, but its shift gain is only 2.46 and has no calibrated significance.



Wide family: $\Delta = 29.19$ MeV and a 2021 offset of $+2.82$ MeV. The offset gain is 3.32; releasing rates still gains 9.10. The fitted offset is close to the allowed 3 MeV scale, so detector calibration would be essential to a physical interpretation.

33. Validation, reproducibility, and references

The numerical audit records 76 passing checks. Every observed and toy limit solves the requested CL_s root; the largest recorded error is 2.12×10^{-8} . Selected full-rank covariance calculations agree with the compressed calculation. The generalized pair fitter reproduces the parent likelihoods; the separate regional replay covers 100 parent rows with a maximum Q_2 difference of 4.1×10^{-12} .

The one-signal solver was independently checked against ten closed one-bin solutions, both bounded-tail branches, zero counts, analytic derivatives, and native parent fits. The shared-separation review checks nesting, gradients, neighboring mass points, and reconstructed count expectations. The injection audit verifies saved count superposition and hashes, positive reference components, the exact trial budget, frozen conversion, and complete-spectrum reuse.

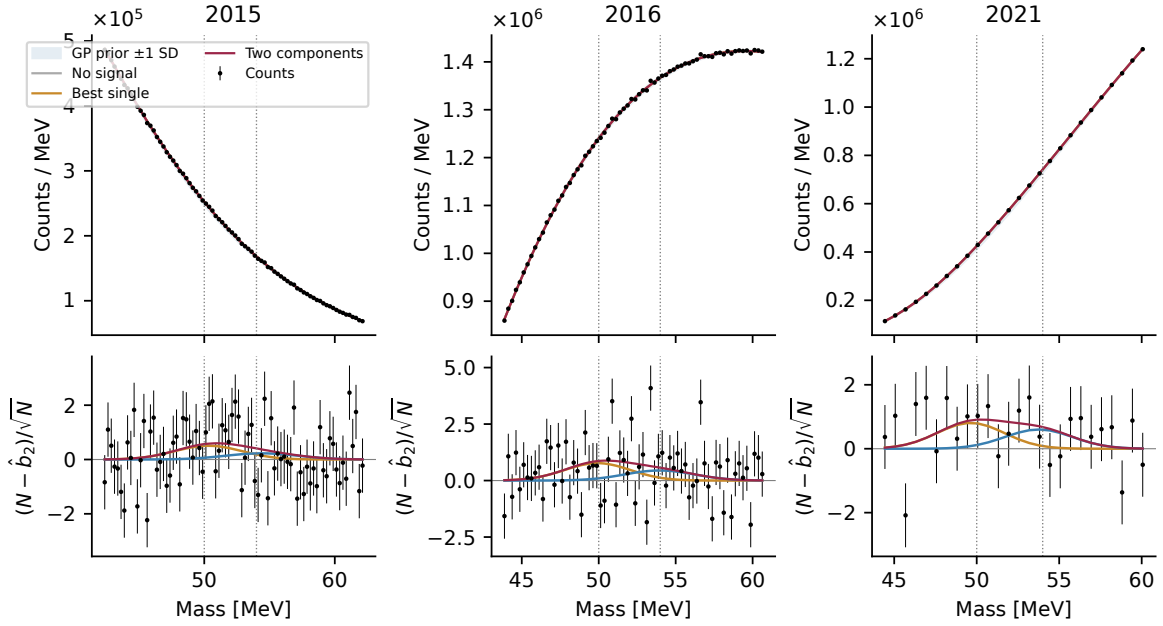
The source package records all spectra and kernel states, parent references, count seeds and increments, 100 trial labels, 90 unique count hashes, fit tables, plot sources, and numerical reviews. Existing cached injection curves are tied to source, input, and output checksums. Rebuilding the PDF requires only the supplied figures and LaTeX source; regenerating numerical products uses Python with NumPy, SciPy, pandas, and Matplotlib. The original ROOT extraction and released-state provenance remain pinned in the input manifest.

The ten trials do not calibrate a look-elsewhere effect, coverage, or a particle-model probability. The strongest follow-up is to test whether the selected spacings and relative rates persist under independent background constraints and measured detector calibration shifts. The present work supplies explicit models, observed scans, and reproducible shape experiments for that comparison.

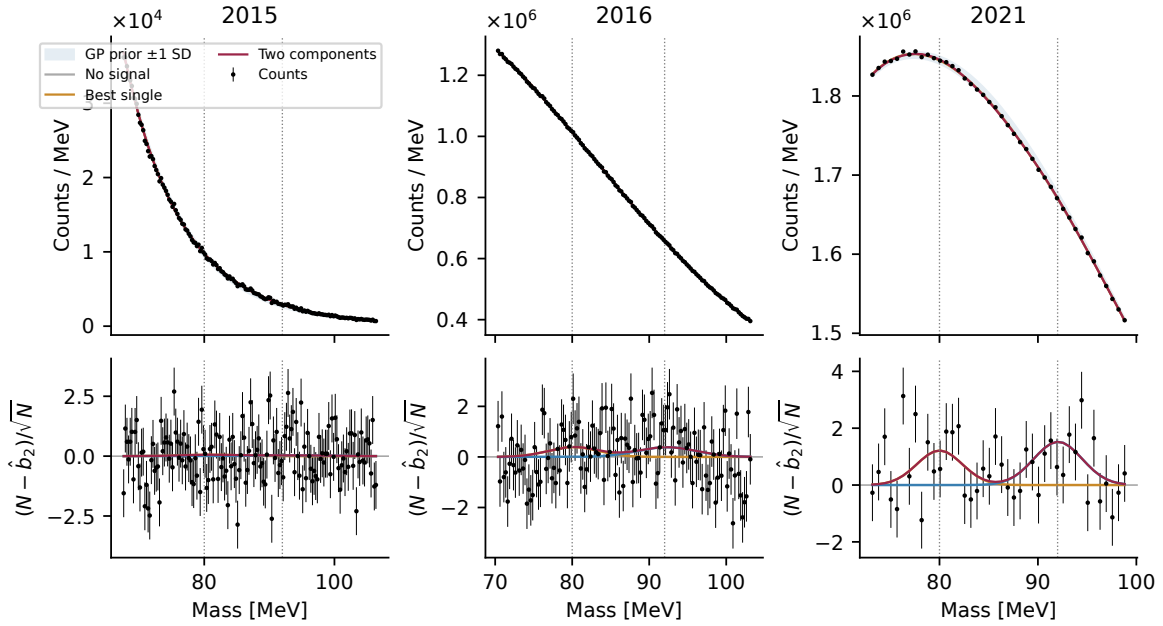
References

- [1] HPS GPR signal-assessment plan, Section 5; and *Two positive components and the intervening dip*, v5.2.0 (10 September 2026). Frozen plan and parent numerical products are included.
- [2] HPS GPR v5.0.2 audited analysis note and significance comparison (9 September 2026), pinned source PDF.
- [3] HPS GPR v5.1.0 independent binning/significance study (9 September 2026), pinned source PDF.
- [4] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C71 (2011)1554, especially the bounded statistic and piecewise tails. [arXiv:1007.1727](#).
- [5] M. Frate et al., *Modeling Smooth Backgrounds and Generic Localized Signals with Gaussian Processes* (2017). [arXiv:1709.05681](#).
- [6] The fixed-pair chi-bar reference and its efficient-information convention follow the derivation and sources documented in the v5.2.0 report. No reference mixture is used for the selected pair-versus-single or mass-shift comparisons.

34. Additional regional extractions

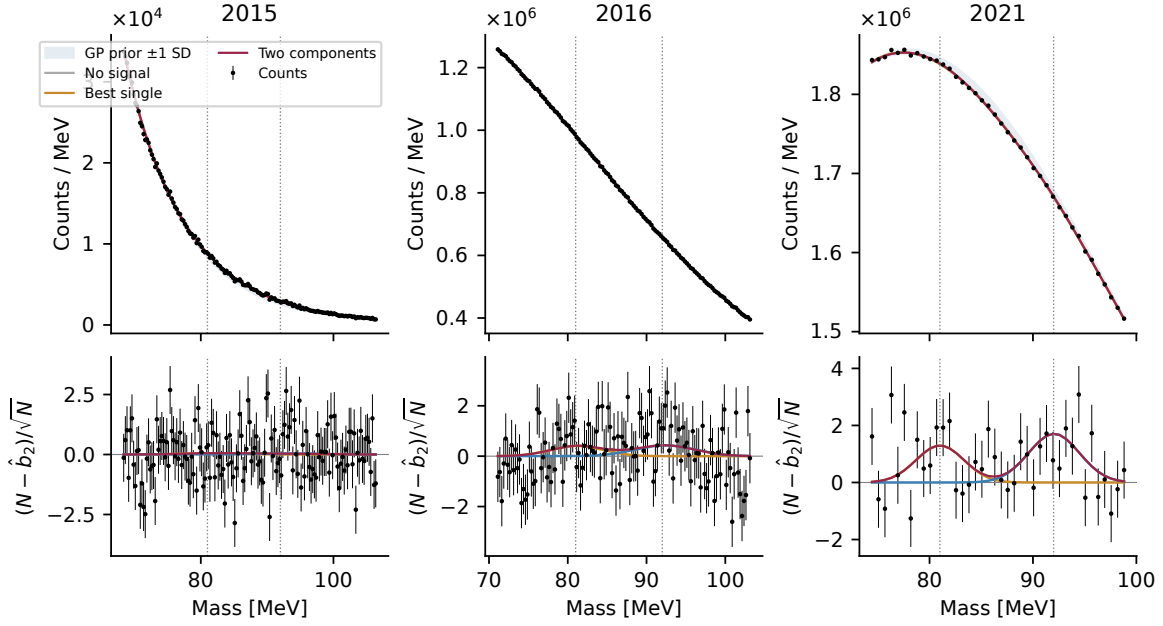


All three: 50/54 MeV; $\Delta_{2/1}^{\text{signed}} = 0.78$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

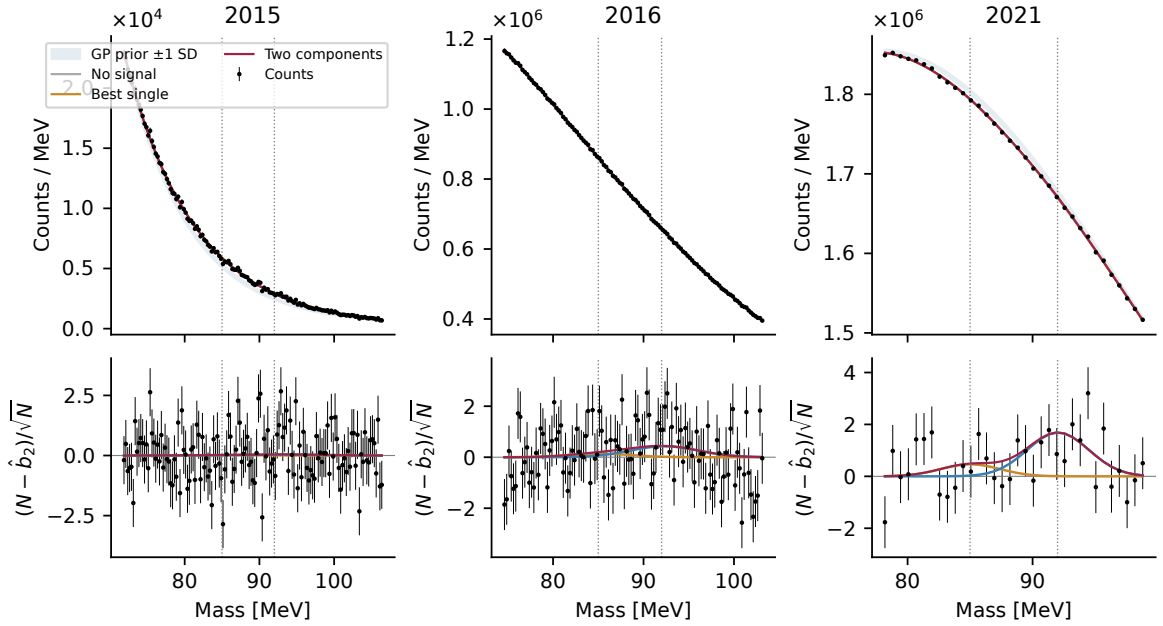


All three: 80/92 MeV; $\Delta_{2/1}^{\text{signed}} = 3.62$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

35. Additional regional extractions (continued)

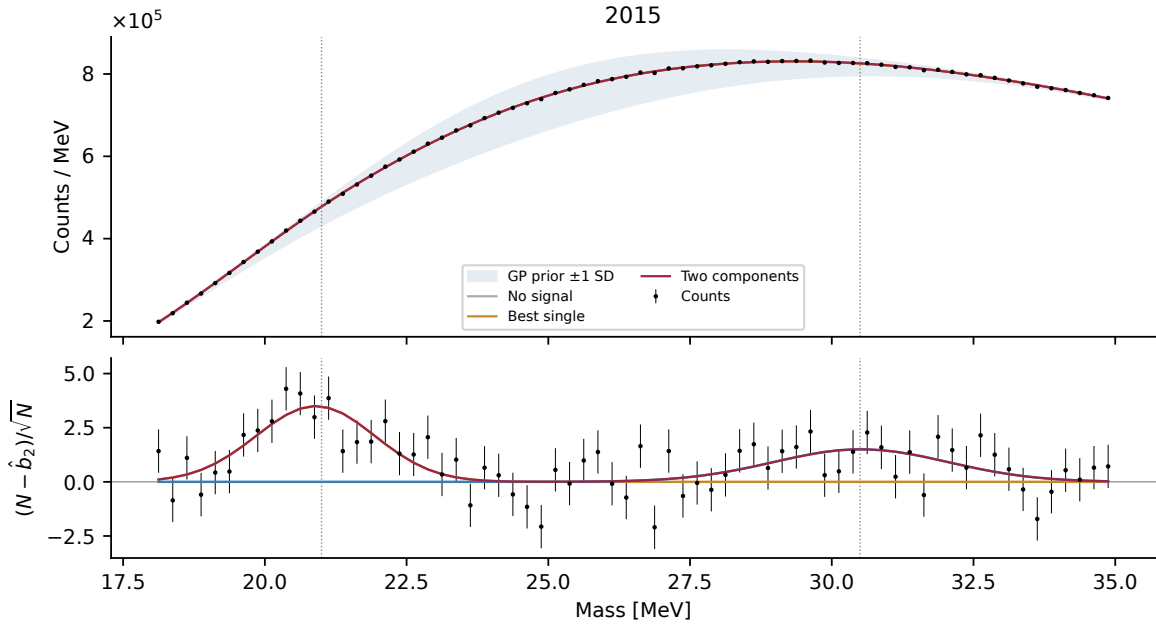


All three: 81/92 MeV; $\Delta_{2/1}^{\text{signed}} = 3.99$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

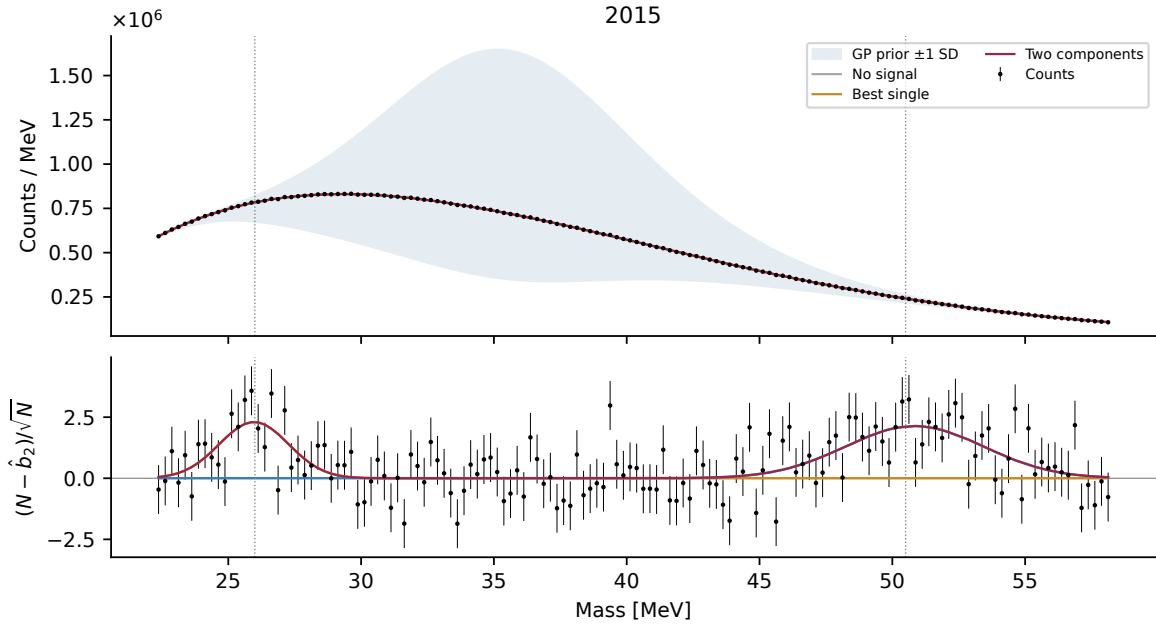


All three: 85/92 MeV; $\Delta_{2/1}^{\text{signed}} = 0.04$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

36. Additional regional extractions (continued)

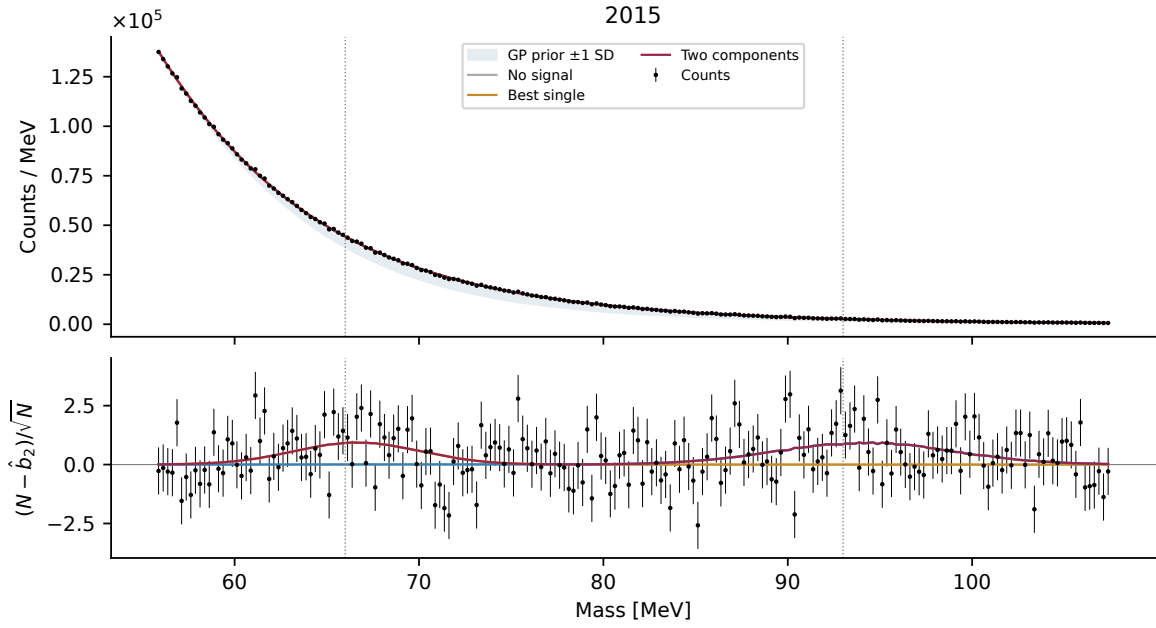


2015: 21/30.5 MeV; $\Delta_{2/1}^{\text{signed}} = 3.17$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

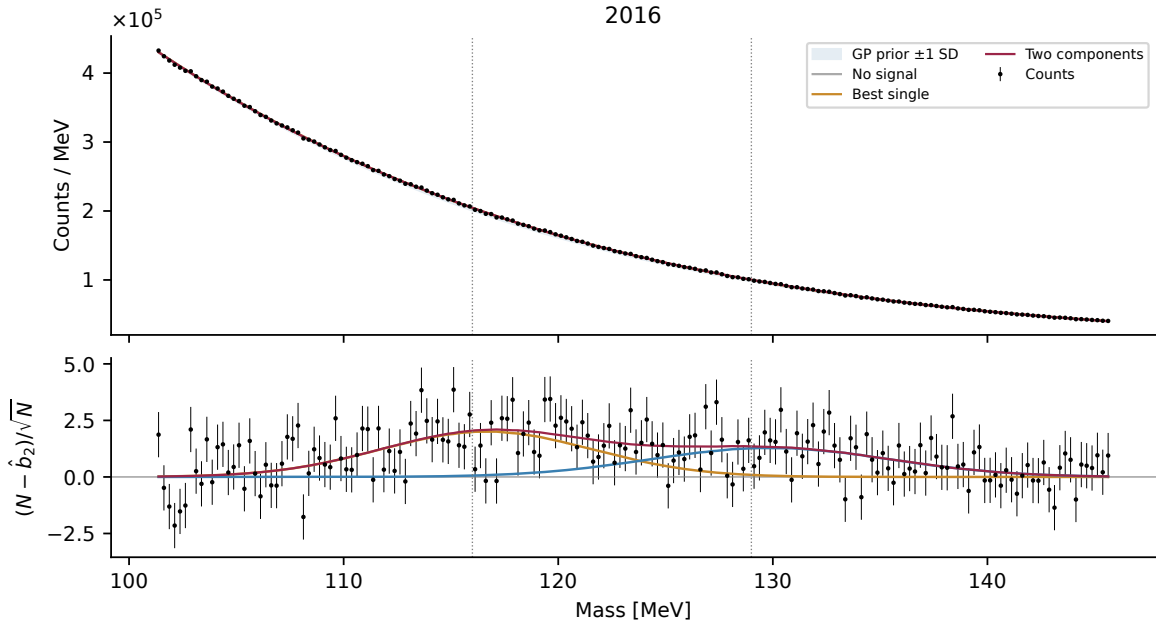


2015: 26/50.5 MeV; $\Delta_{2/1}^{\text{signed}} = 3.17$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

37. Additional regional extractions (continued)

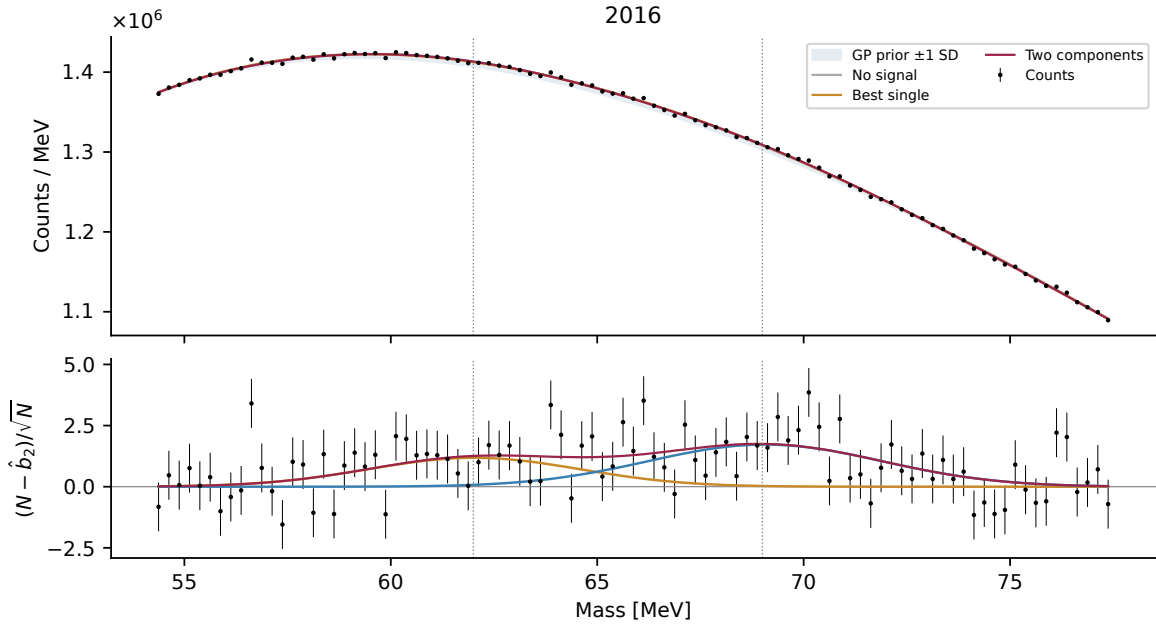


2015: 66/93 MeV; $\Delta_{2/1}^{\text{signed}} = 2.29$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

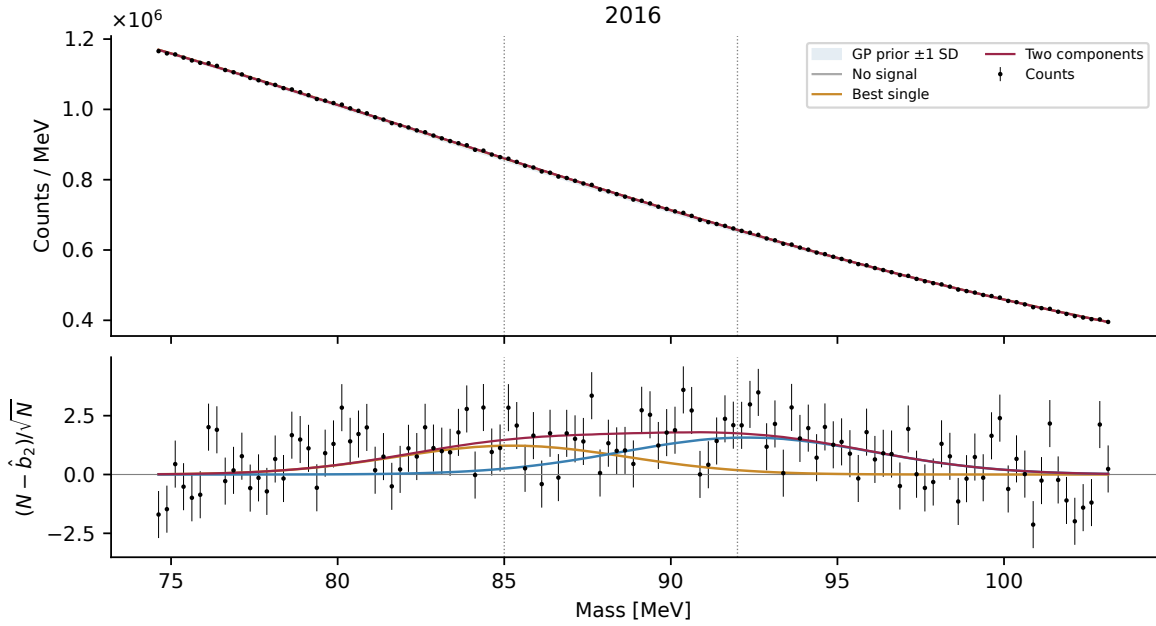


2016: 116/129 MeV; $\Delta_{2/1}^{\text{signed}} = 10.81$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

38. Additional regional extractions (continued)

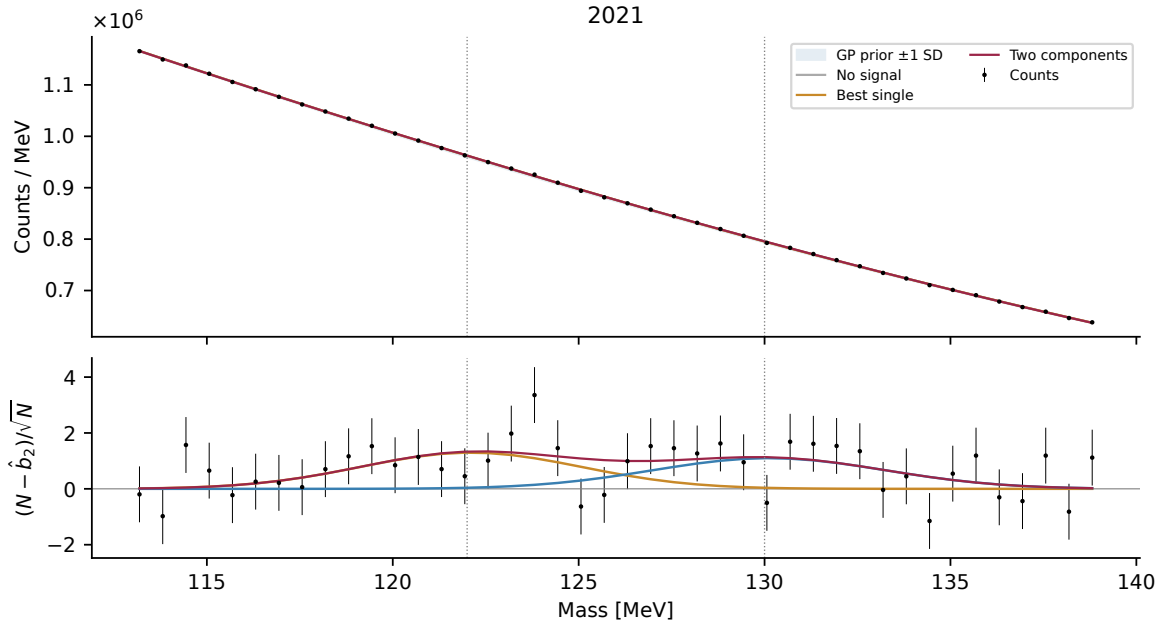


2016: 62/69 MeV; $\Delta_{2/1}^{\text{signed}} = 2.79$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

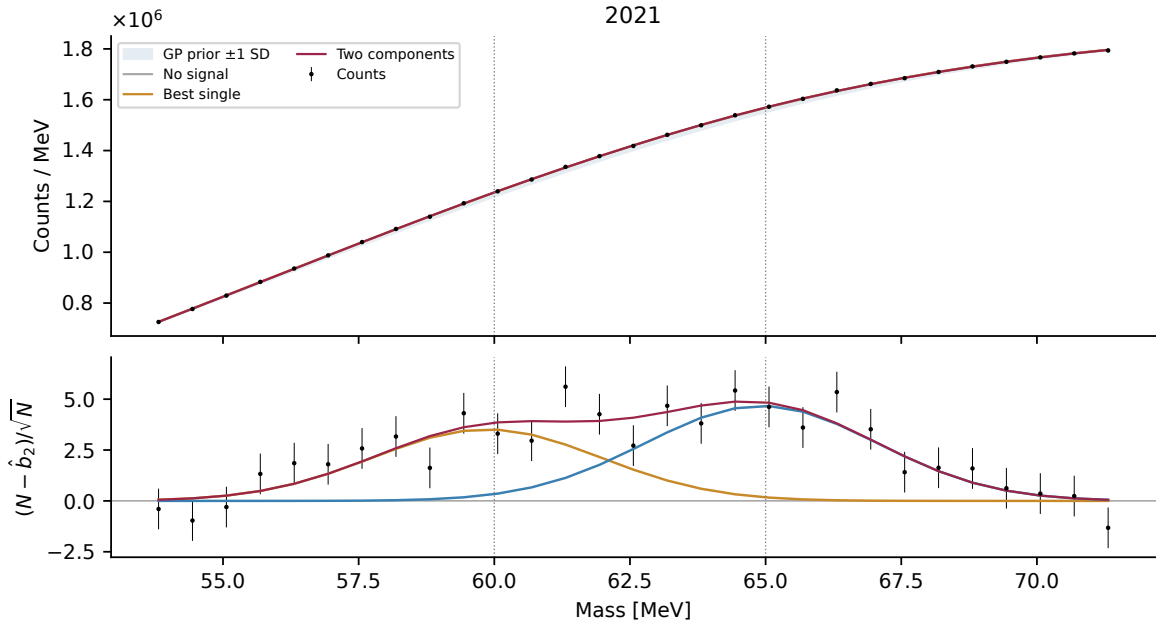


2016: 85/92 MeV; $\Delta_{2/1}^{\text{signed}} = 6.83$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

39. Additional regional extractions (continued)

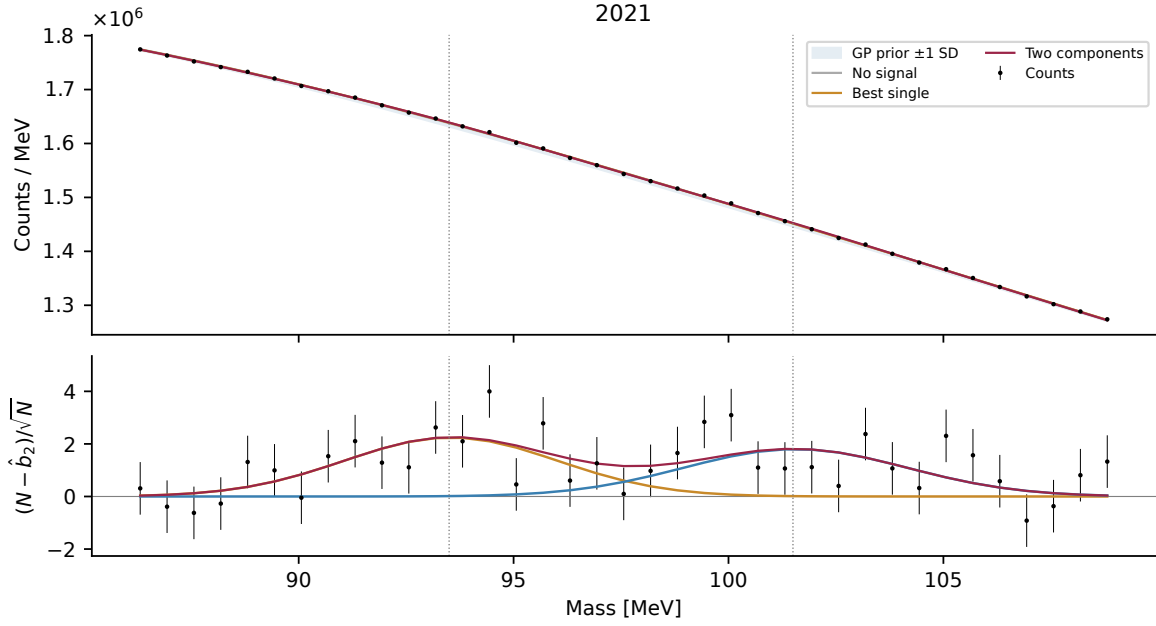


2021 (10%): 122/130 MeV; $\Delta_{2/1}^{\text{signed}} = 2.72$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

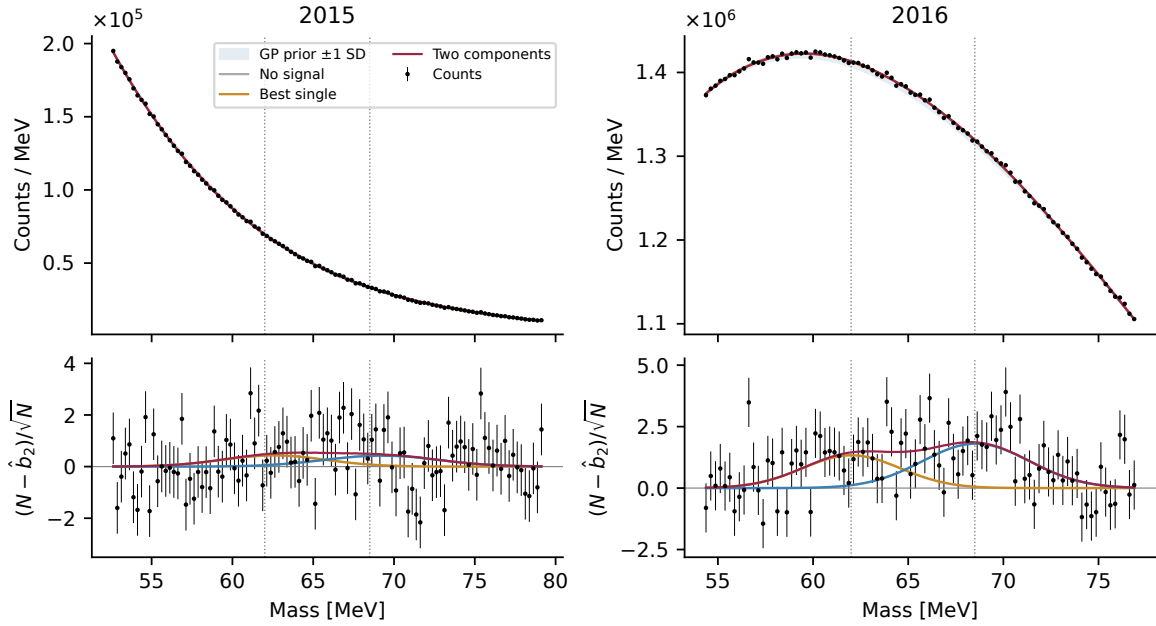


2021 (10%): 60/65 MeV; $\Delta_{2/1}^{\text{signed}} = 5.34$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

40. Additional regional extractions (continued)

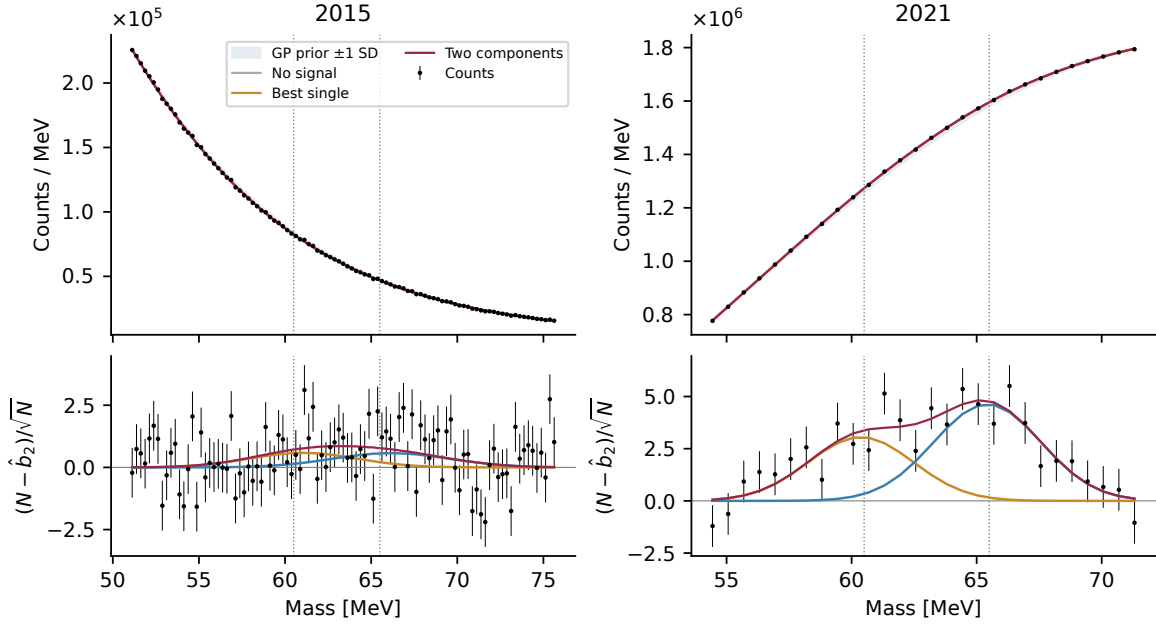


2021 (10%): 93.5/101.5 MeV; $\Delta_{2/1}^{\text{signed}} = 4.84$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

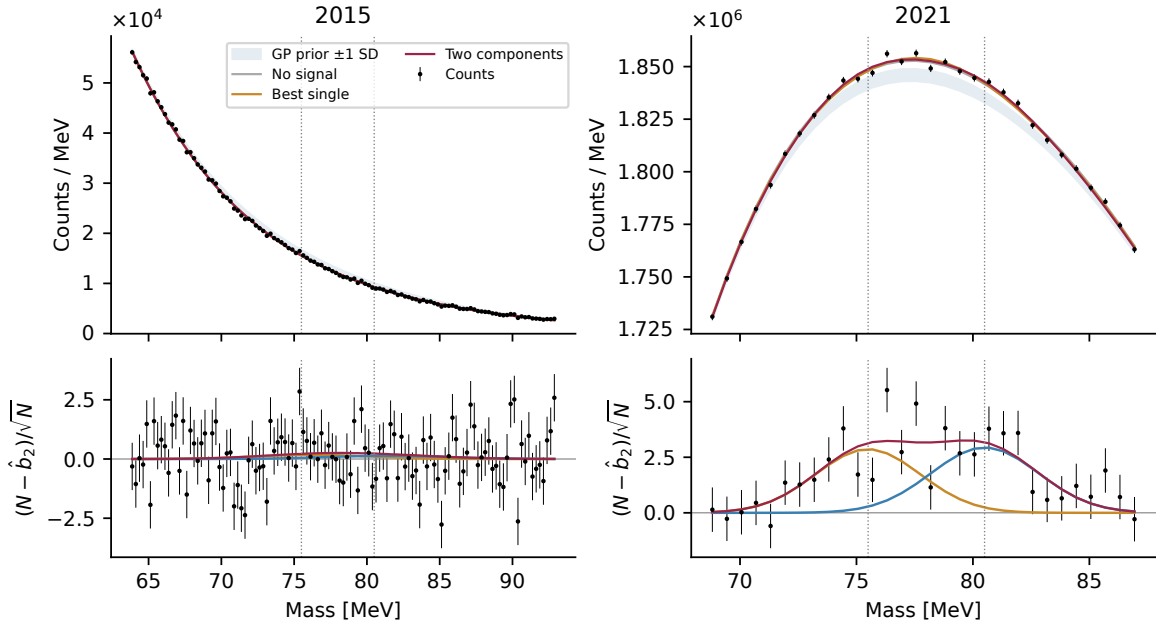


2015 + 2016: 62/68.5 MeV; $\Delta_{2/1}^{\text{signed}} = 4.40$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

41. Additional regional extractions (continued)

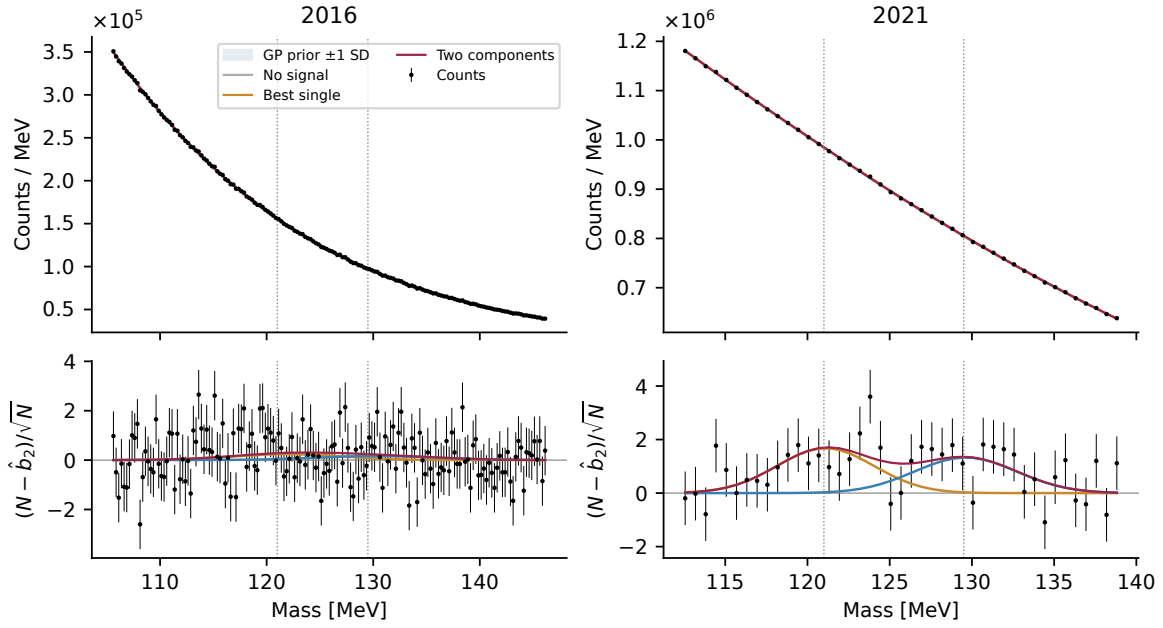


2015 + 2021: 60.5/65.5 MeV; $\Delta_{2/1}^{\text{signed}} = 5.06$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

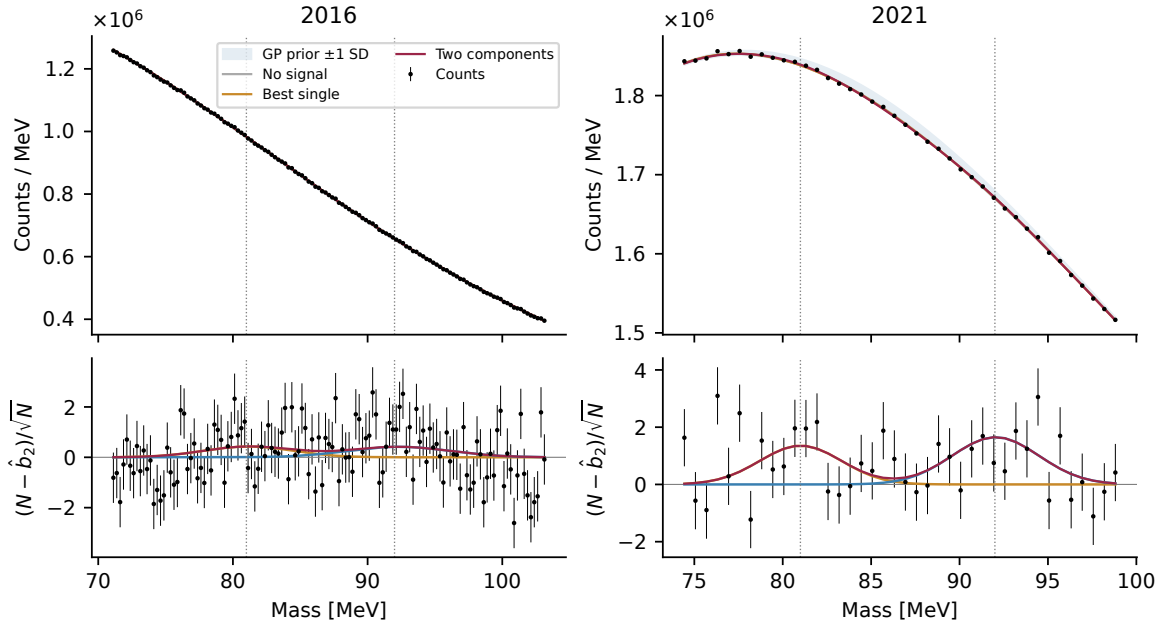


2015 + 2021: 75.5/80.5 MeV; $\Delta_{2/1}^{\text{signed}} = 6.76$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.

42. Additional regional extractions (continued)



2016 + 2021: 121/129.5 MeV; $\Delta_{2/1}^{\text{signed}} = 4.09$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.



2016 + 2021: 81/92 MeV; $\Delta_{2/1}^{\text{signed}} = 4.37$. The same GP constraint and count bins enter no-signal, best-single, and pair fits. Background uncertainty remains in the likelihood; the lower display shows count errors only.